

# Valley Coastal Bend Health Care System

## STATEMENT OF WORK

For

### SPS Deinstallation (old) Cart Washer, Replacement Electric Cart Washer (new), and Installation of Electric Cart Washer (new)

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#### -- Services/Replacement Cart Washer--

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##### **Provide:**

1. Professional services to include but not limited to:
  - a. Deinstallation of Steam Cart Washer
  - b. Pre-installation Site Preparation
  - c. Installation of Electric Cart Washer
  - d. Safety, Testing Completion
2. Replacement Electric Cart Washer

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#### -- Documentation and Acronyms --

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##### **Applicable Documents:**

3. The following laws, regulations, policies, and procedures in effect on date of contract issuance and all subsequent changes or updates apply:
  - a. Federal Acquisition Regulation
  - b. Veterans Administration Acquisition Regulation
  - c. OSHA Standards: [Law and Regulations | Occupational Safety and Health Administration](#)
  - d. Master Construction Specification: <https://www.cfm.va.gov/TIL/spec.asp#03>
  - e. TIL Telecommunications Systems Infrastructure Standards  
<https://www.cfm.va.gov/til/dguide/OIT-InfrastrucStdTelecomSpaces-v4-0.pdf>
4. Additional resources:
  - a. EM 385-1-1 (2014) – U.S. Army Corps of Engineers Safety and Health Requirements

##### **Acronyms:**

5. Glossary: Acronyms used in this SOW are listed below for easy reference:
  - a. CO Contracting Officer
  - b. COR Contracting Officer's Representative
  - c. FAR Federal Acquisition Regulation
  - d. CFM Construction & Facilities Management
  - e. VA Veteran Affairs
  - f. EPA Environmental Protection Agency
  - g. PPE Personal Protective Equipment
  - h. SOW Statement of Work
  - i. OSHA Occupational Safety and Health Administration
  - j. N/A Not Available
  - k. NEC National Electrical Code
  - l. NTP Notice to Proceed
  - m. RFI Request for Information

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**-- Project Scope --**

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**GENERAL:**

6. Valley Coastal Bend Healthcare System (VCB HCS) intends to procure an Electric Cart Washer to replace the existing Steam Cart Washer. The current Steam Cart Washer is non-functional and the facility no longer has the infrastructure capabilities for operation.
7. This contract will include the deinstallation of existing Cart Washer, pre-site preparation work, installation of new Electric Cart Washer, and final testing of functionality of equipment.
8. The contract will be a total "Turn-Key" project upon completion.
  - a. The Contractor shall provide all labor, material, equipment, and supervision required to complete the work.
  - b. The Contractor shall ensure that all work meets applicable local & state code and standards.

**SPECIFICS (services):**

9. Deinstallation of Steam Cart Washer
  - a. Contractor shall safely remove existing stainless-steel panels around the existing Steam Cart Washer.
  - b. Contractor shall safely remove the existing Steam Cart Washer
  - c. Contractor shall dispose of existing Steam Cart Washer in a dumpster provided by the Contractor at the rear of the facility.
  - d. Contractor shall furnish labor to demo/remove existing utilities that do not meet new Electric Cart Washer specifications.
  - e. Contractor shall make adjustments to service elevator if needed for installation and deinstallation and return to normal compacity upon completion.
  - f. Contractor shall provide dumpster for the disposal of old Steam Cart Washer
  - g. Contractor shall provide forklift for the removal and disposal of Steam Cart Washer in dumpster
10. Pre-Installation Site Preparation.
  - a. Contractor shall furnish labor and materials to drill/core concrete for utilities required for new Cart Washer equipment.
  - b. Contractor shall furnish labor and materials to install new stainless steel cart washer.
  - c. Contractor shall furnish labor and materials to install new structural support within electrical room for the purpose of the Cart Washer.
  - d. Contractor shall install step-down transformer in an elevated position above existing facilities electrical transformer due to limited space.
  - e. Contractor shall ensure thermal moisture protection to include fire caulking though fire walls and fire caulking all coring penetrations through concrete.
  - f. Contractor shall furnish labor and materials to remove/reinstall designated 3rd floor window to accommodate the ICRA exhaust route.
  - g. The Contractor shall demo/remove any utility water (hot and cold) connections, R.O system connections, compressed air system connections, and drainage system connections that do not comply with new Electric Cart Washer specifications.
  - h. Contractor shall connect or create connections for utility water (hot and cold), R.O system, compressed air system, drainage system and any final connections to new Electric Cart Washer.
  - i. The Contractor shall connect the Cart Washer to the existing water supply and drainage system, ensuring all connections are secured and meet applicable plumbing codes.
  - j. The Contractor shall test the new Cart Washer to verify its performance and functionality.
  - k. Contractor will be responsible for any and all unforeseen plumbing and electrical repairs encountered during install and removal.

11. Installation of Electric Cart Washer:

- a. Contractor shall connect or create connections for utility water (hot and cold), R.O system, compressed air system, drainage system and any final connections to new Electric Cart Washer.
- b. Contractor shall install an exhaust fan and integrate it into the existing duct system to meet cart washer specifications.
- c. Contractor shall install and connect all plumbing to current RO system, including installing valves to separate new lines from existing ones.
- d. Contractor shall install a critical water Booster above the cart washer.
- e. Contractor shall provide forklift to unload/unbox new Electric Cart Washer.
- f. Contractor shall provide crane/man-lift for removal of 3<sup>rd</sup> floor window (removal of 3<sup>rd</sup> floor window to accommodate ICRA exhaust route).
- g. Contractor shall furnish labor and materials to install new structural support within electrical room for the purpose of step-down transformer in an elevated position above existing facilities electrical transformer due to limited space.
- h. Contractor shall provide labor and material for all LOTO protocols.
- i. Contractor shall provide labor and materials to disconnect existing Steam Cart Washer electrical.
- j. Contractor shall provide labor and materials to install new 480v/100a breaker to existing panel within designated electrical room.
- k. Contractor shall provide labor and materials to install new conduits, conductors, boxes, fusible disconnects as required by manufacturer in accordance with National Electric Code.
- l. Any electrical shutdowns will be conducted after working hours in coordination and approval by COR.
- m. Contractor to provide labor and materials to make final connections to new Electric Cart Washer.
- n. All persons performing electrical work under this contract shall be licensed journeyman or master electricians. All apprentice electricians performing under this contract shall be deemed unqualified persons unless they are working under the immediate supervision of a licensed electrician or master electrician.

12. Safety, Testing, Completion

- a. The Contractor shall provide all PPE required for the employees and sub-contracting employees associated with this work.
- b. The Contractor shall adhere to all safety and Infection Control Measures Level IV while performing work at the facility (see attachment ICRA/PCRA).
- c. The Contractor shall provide their own dumpster and dispose of all damaged items, removed items, and debris created from the construction and demolition in every aspect of this project.
- d. The Contractor shall adhere to all local VA Policies.

**Specifics: Replacement Equipment:**

13. Contractor shall supply Electric Cart Washer with at minimum the following salient characteristics:

- i. Max. dimensions of cart washer overall: L 10' x W 10' x H 11'
- ii. Fully electrical boiler/electric heated
- iii. Transparent (tempered glass or similar) double door pass through at each end of the chamber
- iv. Power operated chamber doors that provide a positive seal to prevent air and water leakage.
- v. Full size loading ramp
- vi. Forced air drying
- vii. Non-recirculating drying air to prevent cross contamination

- viii. Oscillating jet-spray manifolds
- ix. Critical water DIW (Deionized Water) valve
- x. Critical water booster pump
- xi. Reuse tank of 150L minimum.
- xii. Automatic chemical dispensing for 3 types of detergents. (enzymatic, non-enzymatic, lubricant)
- xiii. Automatic chemical dispensing pump flowmeter.
- xiv. Drain tempering/drain cooling
- xv. Booster heater
- xvi. Chamber guardrails
- xvii. Loading/unloading ramps
- xviii. Drain pan
- xix. Tilting mechanism to allow water run off
- xx. Audible and visual alarm kit to indicate cycle completion
- xxi. Electrical transformer for electric heat (480V/400V, 80KVA)
- xxii. Lid stand x2
- xxiii. Washer racks for containers
- xxiv. Washer racks for instruments
- xxv. Energy efficient heat exchanger
- xxvi. Building exhaust connection
- xxvii. Debris filter
- xxviii. Training to technicians for operation and maintenance

**Other Specifics:**

- 14. The Contractor shall provide all PPE required for the employees and sub-contracting employees associated with this work area. (Hair Covers, Beard Covers, Disposable Coveralls, Covers must be worn when entering this work and removed upon leaving area. PPE cannot be worn in hallway.)
- 15. The Contractor shall provide to the COR a developed work plan and schedule detailing, of at a minimum, the procedures to be employed, the equipment and materials to be used, the interim life safety measure to be used during the work, and a schedule defining the duration of the work with milestone subtasks to include at the minimum; the deinstallation, site preparation work, installation, and completion of the project.
- 16. Contractor shall provide critical water testing upon completion according to AMMI ST 108 requirements and VCB policy.
- 17. Contractor is responsible to patch, repair, or replace all finishes to include drywall, painted walls, floor tiles, and crash bars/rails if altered/damaged from its original state prior to completion.
- 18. The Contractor shall adhere to all safety and infection control measures while performing work at the facility.
- 19. The Contractor shall provide their own dumpster and dispose of all damaged items, removed items, and debris created from the construction and demolition in every aspect of this project.
- 20. The Contractor shall adhere to all local VA Policies.
- 21. The Contractor shall meet ICRA **IV** requirements provide in appendix D.

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**-- Site Investigation --**

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**Expectation:** Contractors shall determine existing as-built conditions through site visits. The Contractor shall submit any alternate suggestions for required materials for approval prior to procurement.

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**-- Environmental/Occupational Safety and health (EOSH) --**

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**Expectation:** All projects and project designs shall contain language that addresses all applicable EOSH requirements.

22. Regulatory Compliance- EOSH requirements for work performed at the above location can be found in:
- a. Volume 29 of the Code of Federal Regulations, Part 1910- Occupational Safety for General industry.
  - b. Volume 29 of the Code of Federal Regulations, Part 1926- Occupational Safety for the Construction Industry.
  - c. Volume 40 of Code of Federal Regulations, Protecting of the Environment.
  - d. Volume 49 of the Code of Federal Regulations, Part 397- Transportation of Hazardous Materials.

**Specification:**

23. Additional Appendix includes:
- a. Safety Requirements
  - b. General Requirements
  - c. AAMI ST 108
  - d. PCRA
  - e. ICRA

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**-- Location/Hours/Deliverables --**

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**Performance Location:**

24. Harlingen Care Center located at 2601 Veterans Dr., Harlingen, TX, 78550 as required by the Texas Valley Coastal Bend Health Care System.

**Hours:**

25. Demolition and electrical shutdowns are to be performed outside of normal operation hours (Normal Hours are Monday to Friday between 07:00AM - 08:00PM Central Standard Time). Weekend work is allowed with coordination and approval from COR.

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**-- Contractor Personnel and Management --**

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**Contractor Personnel:**

26. The Contractor shall be responsible for employing qualified personnel to perform the work specified in this SOW. The Contractor shall maintain the personnel, organization and administrative control necessary to ensure that the work delivered meets the contract specifications and requirements.
27. The Contractor is required to ensure that all subcontractor or employees have been screened by the VCB Police. All personnel are required to wear a VA government furnished badge. All contractors and subcontractors are required to wear a government and company furnished badge while performing services under this contract. The badge will contain a personal picture, name of employee, and Contractor's name. Badges shall be worn on the outer garment attached to the outer shirt or jacket pocket by a button or clip or worn around the neck secured by an appropriate identification card lanyard.
28. Government issued identification badge remains the property of the Government and shall be returned to the issuing office or other designated Government personnel upon transfer of Contractor employees from the contract.

**Management:**

29. All information generated in performance of this contract becomes property of the Government and must be surrendered upon request and/or completion of the contract. The Contractor must protect against the destruction and/or misplacement of all information produced under this contract, unless directed under local policy to destroy. This includes both hard and soft copies of data .

30. The Contractor shall not release any information (including photographs, files, public announcements, statements, denials, or confirmations) on any part of the subject matter of this contract or any phase of any program hereunder without the prior written approval of the COR.
31. Contractor personnel or subcontractor shall conduct themselves in a professional manner (i.e., timeliness, communication; spoken and written etc.). The Contracting Officer may require the Contractor to remove from the job site any employee working under this contract for reasons of suspected misconduct, a suspected security breach, or suspected to be under the influence of alcohol, drugs, or any other incapacitating agent. Contractor employees shall be subject to dismissal from the premises upon determination by the Contracting Officer that such action is necessary in the interests of the Government. The hospital director or designated representative has the authority to bar individuals from the Government facility. The removal from the job site or dismissal from the premises shall not relieve the Contractor of the requirement to provide sufficient personnel to perform the services as required by this performance work statement. The Government shall not reimburse the Contractor for travel and other expenses associated with the removal of personnel.
32. Contractor personnel are not authorized to carry or possess personal weapons to include, but not limited to, firearms and knives with a blade length in excess of three inches, while assigned under this contract.

**Safety/Security Requirements:**

33. The Contractor shall comply with all applicable federal, state and local laws and ensure performance is secure while protecting material, equipment, and supplies from damage and loss. The Contractor shall ensure that employees or subcontractor have the proper PPE for the job. Government security personnel shall have the express right to inspect for security violations at any time during the term of the contract.
34. The VCB HCS Safety Officer and COR shall have full authority to see that the contractor obeys all safety rules and regulations relative to the fulfillment of this contract.

**Work Attire and Appearance:**

35. Contractor personnel shall present a professional appearance commensurate with standards delineated for Government civilian personnel acting in similar capacities.
36. The Contractor shall provide the necessary PPE to perform all the work described in this SOW.

**Personnel Qualification:**

37. The Contractor shall be responsible for employing technically qualified personnel to perform the work specified in this SOW and authorized to work in the United States. The Contractor shall maintain the personnel, organization, and administrative control necessary to ensure that the work delivered meets the contract specifications and requirements. The work history of each contractor employee must contain experience directly related to the task and functions he/she is intended to perform under this contract. The Government reserves the right, during the life of the resulting contract, to request work histories on any contractor employee for the purposes of verifying compliance with the above requirements; additionally, the Government reserves the right to review resumes of contractor personnel proposed to be assigned. Personnel assigned to, or utilized by, the Contractor in performance of work shall be fully capable of performing the requirements contained in the SOW in an efficient, reliable, and professional manner.
38. Contractor shall provide a supervisor who speaks English on site at all times.
39. Personnel should be properly trained for safety and performance of their respective duties.
40. The Government reserves the right to review Contractor personnel qualifications. Upon request, resume shall be provided to the Contracting Office prior to clearance approval, for review and acceptance by the COR. The intent is verification purpose and not for the Government to become the hiring authority.
41. Contractor and all contractor employees must have taken the Occupational Safety and Health Administration (OSHA) Certified 30/ 10-hour construction safety course and show documented proof. The contractor shall designate a "competent person" in accordance with 29 CFR 1926. 32(f) which states, "One who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous or dangerous to employees and who has authorization to take prompt corrective measures to eliminate them". The contractor shall instruct each of his employees in the recognition and avoidance of unsafe conditions and the

regulations applicable to this work environment to control or eliminate any hazards or other exposure to illness or injury.

**Non-Personal Service Statement:**

42. Contractor employees performing services under this order will be controlled, directed, and supervised at all times by management personnel of the contractor. Contractor management will ensure that employees properly comply with the performance work standards outlined in the SOW. Contractor employees will perform their duties independent of, and without the supervision of, any Government official. The tasks, duties, and responsibilities set forth in the contract may not be interpreted or implemented in any manner that results in any contractor employee creating or modifying Federal policy, obligating the appropriated funds of the United States Government, overseeing the work of Federal employees, providing direct personal services to any Federal employee, or otherwise violating the prohibitions set forth in Parts 7.5 and 37.1 of the Federal Acquisition Regulation (FAR). The Government will control access to the facility and will perform the inspection and acceptance of the completed work.

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**-- Damages/Insurance --**

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**Damages:**

43. The Contractor is liable for all damages to Government equipment, property; utilities, etc., located at the site (e.g., sprinkler heads, ceiling tiles, manholes, fire hydrants, vehicles, windows, etc.) caused by the Contractor, its employees or its subcontractors. The Contractor shall immediately report all damages to the COR and shall provide the Contracting Officer (CO) and COR, a written report, within 24 hrs. The report shall include **What occurred, who was involved, when it occurred (Time/Date), where did it occurred, why it occurred, how it occurred, and the proposed means to repair or replace.**
44. Contractor is to maintain free and unobstructed access to facility emergency services and for fire, police and other emergency response forces in accordance with NFPA 241.

# APPENDIX A

## SAFETY REQUIREMENTS

### ACCIDENT PREVENTION PLAN (APP) :

- A. The APP (aka Construction Safety & Health Plan) shall interface with the Contractor's overall safety and health program. Include any portions of the Contractor's overall safety and health program referenced in the APP in the applicable APP element and ensure it is site-specific. The Government considers the Prime Contractor to be the "controlling authority" for all worksite safety and health of each subcontractor(s). Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that Mishap prevention responsibilities are being carried out.
- B. The APP shall be prepared as follows:
1. Written in English by a qualified person who is employed by the Prime Contractor articulating the specific work and hazards pertaining to the contract. Specifically articulating the safety requirements found within these VA contract safety specifications and the latest version of the United States Army Corps of Engineers - Safety and Occupational Health (SOH) Requirements - EM 385-1-1



Manual. Model language and format can be found in Appendix A of the EM 385-1-1 Manual

2. Address both the Prime Contractors and the subcontractors work operations.
3. State measures to be taken to control hazards associated with materials, services, or equipment provided by suppliers.
4. Address all the elements/sub-elements and in order as follows:
  - a. **SIGNATURE SHEET.** Title, signature, and phone number of the following:
    - 1) Plan preparer (Qualified Person such as corporate safety staff person or contracted Certified Safety Professional with construction safety experience);
    - 2) Plan approver (company/corporate officers authorized to obligate the company);
    - 3) Plan concurrence (e.g., Chief of Operations, Corporate Chief of Safety, Corporate Industrial Hygienist, project manager or superintendent, project safety professional). Provide concurrence of other applicable corporate and project personnel (Contractor).
  - b. **BACKGROUND INFORMATION.** List the following:
    - 1) Contractor;
    - 2) Contract number;
    - 3) Project name;

4) Brief project description, description of work to be performed, and location; phases of work anticipated (these will require an AHA).

c. **STATEMENT OF SAFETY AND HEALTH POLICY.** Provide a copy of current corporate/company Safety and Health Policy Statement, detailing commitment to providing a safe and healthful workplace for all employees. The Contractor's written safety program goals, objectives, and accident experience goals for this contract should be provided. The Statement of Safety and Health Policy must be signed by a company executive.

d. **RESPONSIBILITIES AND LINES OF AUTHORITIES.** Provide the following:

- 1) A statement of the employer's ultimate responsibility for the implementation of his SOH program;
- 2) Identification and accountability of personnel responsible for safety at both corporate and project level. Contracts specifically requiring safety or industrial hygiene personnel shall include a copy of their resumes.
- 3) The names of Competent and/or Qualified Person(s) and proof of competency/qualification to meet specific OSHA or EM 385-1-1 Competent/Qualified Person(s) requirements must be attached.;
- 4) Requirements that no work shall be performed unless a designated competent person is present on the job site;

5) Requirements for pre-task Activity Hazard Analysis (AHAs);

6) Lines of authority;

7) Policies and procedures regarding noncompliance with safety requirements (to include disciplinary actions for violation of safety requirements) should be identified;

e. **SUBCONTRACTORS AND SUPPLIERS.** If applicable, provide procedures for coordinating SOH activities with other employers on the job site:

1) Identification of subcontractors and suppliers (if known);

2) Safety responsibilities of subcontractors and suppliers.

f. **TRAINING.**

1) Site-specific SOH orientation training at the time of initial hire or assignment to the project for every employee before working on the project site is required.

2) Mandatory training and certifications that are applicable to this project (e.g., explosive actuated tools, crane operator, rigger, crane signal person, fall protection, electrical lockout/NFPA 70E, machine/equipment lockout, confined space, etc....) and any requirements for periodic retraining/recertification are required.

- 3) Procedures for ongoing safety and health training for supervisors and employees shall be established to address changes in site hazards/conditions.
- 4) OSHA 10-hour Construction Outreach training within the past five years is required for all workers on site and the OSHA 30-hour Construction Outreach training within the past five years is required for Trade Competent Persons (CPs)

**g. SAFETY AND HEALTH INSPECTIONS.**

- 1) Specific assignment of responsibilities for a minimum daily job site safety and health inspection during periods of work activity: Who will conduct (e.g., "Site Safety and Health CP"), proof of inspector's training/qualifications, when inspections will be conducted, procedures for documentation, deficiency tracking system, and follow-up procedures.
- 2) Any external inspections/certifications that may be required (e.g., contracted CSP or CSHT)

**h. MISHAP INVESTIGATION & REPORTING.**

- 1) The APP shall include identify person(s) responsible to provide the following to the Contracting Officer Representative:
  - a) Monthly HS&E Performance Report (Exposure Hours Data);
  - b) Mishap investigation reports;
  - c) Project site injury and illness logs;
  - d) Near-Miss reports;

i. **PLANS (PROGRAMS, PROCEDURES) REQUIRED.** Based on a risk assessment of contracted activities and on mandatory OSHA and/or EM 385-1-1 compliance programs, the Contractor shall address all applicable occupational, patient, and public safety risks in site-specific compliance and accident prevention plans. These Plans shall include but are not limited to procedures for addressing the risks associated with the following:

- 1) Emergency response;
- 2) Contingency for severe weather;
- 3) Fire Prevention;
- 4) Medical Support;
- 5) Posting of emergency telephone numbers;
- 6) Prevention of alcohol and drug abuse;
- 7) Site sanitation (housekeeping, drinking water, toilets);
- 8) Night operations and lighting;
- 9) Hazard communication program;
- 10) Welding/Cutting "Hot" work;
- 11) Electrical Safe Work Practices (Electrical LOTO/NFPA 70E);
- 12) General Electrical Safety;
- 13) Hazardous energy control (LOTO);
- 14) Site-Specific Fall Protection & Prevention;
- 15) Excavation/trenching;

- 16) Asbestos abatement;
- 17) Lead abatement;
- 18) Crane Critical lift;
- 19) Respiratory protection;
- 20) Health hazard control program;
- 21) Radiation Safety Program;
- 22) Abrasive blasting;
- 23) Heat/Cold Stress Monitoring;
- 24) Crystalline Silica Monitoring (Assessment);
- 25) Demolition plan (to include engineering survey);
- 26) Formwork and shoring erection and removal;
- 27) Pre-Cast Concrete;
- 28) Public (Mandatory compliance with ANSI/ASSP A10.34-2012);
- 29) Sub-contractor or cross-trade coordination;

C. Submit the APP to the Contracting Officer Representative, Jaime Noyola for review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES 15 calendar days prior to the date of the preconstruction safety conference for acceptance. Work cannot proceed without an accepted APP.

D. Once accepted by the Contracting Officer Representative, the APP and attachments will be enforced as part of the contract. Disregarding the provisions of this contract or the accepted APP will be cause for stopping of work, at the discretion of the Contracting Officer in accordance with

FAR Clause 52.236-13, *Accident Prevention*, until the matter has been rectified.

- E. Once work begins, changes to the accepted APP shall be made with the knowledge and concurrence of the Resident Engineer, Contracting Officer Representative or Government Designated Authority. Should any severe hazard exposure, i.e. imminent danger, become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Resident Engineering Office and Contracting Officer within 2 hours of discovery. Eliminate/remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions to safeguard onsite personnel, visitors, the public and the environment.

**ACTIVITY HAZARD ANALYSES (AHAs) :**

- A. AHAs are also known as Job Hazard Analyses, Job Safety Analyses, and Activity Safety Analyses. Before beginning each work activity involving a type of work presenting hazards not experienced in previous project operations or where a new work crew or sub-contractor is to perform the work, the Contractor(s) performing that work activity shall prepare an AHA (Example electronic AHA forms can be found on the US Army Corps of Engineers web site)
- B. AHAs shall define the activities being performed and identify the work sequences, the specific anticipated hazards, site conditions, equipment, materials, and the control measures to be implemented to eliminate or reduce each hazard to an acceptable level of risk.

C. Work shall not begin until the AHA for the work activity has been accepted by the Contracting Officer Representative and discussed with all engaged in the activity, including the Contractor, subcontractor(s), and Government on-site representatives at preparatory and initial control phase meetings.

1. The names of the Competent/Qualified Person(s) required for a particular activity (for example, excavations, scaffolding, fall protection, other activities as specified by OSHA, EM 385-1-1, or other State and Local agencies) shall be identified and included in the AHA. Certification of their competency/qualification shall be submitted to the Government Designated Authority (GDA) for acceptance prior to the start of that work activity.
2. The AHA shall be reviewed and modified as necessary to address changing site conditions, operations, or change of competent/qualified person(s).
  - a. If more than one Competent/Qualified Person is used on the AHA activity, a list of names shall be submitted as an attachment to the AHA. Those listed must be Competent/Qualified for the type of work involved in the AHA and familiar with current site safety issues.
  - b. If a new Competent/Qualified Person (not on the original list) is added, the list shall be updated (an administrative action not requiring an updated AHA). The new person shall acknowledge in writing that he or she has reviewed the AHA and is familiar with current site safety issues.
3. Submit AHAs to the Contracting Officer Representative, Jaime Noyola, for review for compliance with contract



requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES for review at least 15 calendar days prior to the start of each phase.

Subsequent AHAs as shall be formatted as amendments to the APP. The analysis should be used during daily inspections to ensure the implementation and effectiveness of the activity's safety and health controls.

4. The AHA list will be reviewed periodically (at least weekly) at the Contractor supervisory safety meeting and updated as necessary when procedures, scheduling, or hazards change.
5. Develop the activity hazard analysis using the project schedule as the basis for the activities performed. All activities listed on the project schedule will require an AHA. The AHAs will be developed by the contractor, supplier, or subcontractor and provided to the prime contractor for review and approval and then submitted to the Contracting Officer Representative or Government Designated Authority for review for compliance.

**PRECONSTRUCTION SAFETY CONFERENCE:**

- A. Contractor representatives who have a responsibility or significant role in implementation of the accident prevention program, as required by 29 CFR 1926.20(b)(1), on the project shall attend the pre-construction safety conference to gain a mutual understanding of its implementation. This includes the project superintendent, subcontractor superintendents, and any other assigned safety and health professionals.

B. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the contract. This list of proposed AHAs will be reviewed at the conference and an agreement will be reached between the Contractor and the Contracting Officer's representative as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, review, and acceptance of AHAs to preclude project delays.

**SAFETY MEETINGS:**

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors on the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

**"SITE SAFETY AND HEALTH OFFICER" (SSHO) AND "COMPETENT PERSON" (CP) :**

A. The Prime Contractor shall provide a Safety oversight team that includes a minimum of one (1) person at each project

site, for each shift, to function as the Site Safety and Health Officer (SSHO). The Prime Contractor shall provide a minimum of one "Full-Time" SSHO at each project site, for each shift (with no other duties) that holds as current, a professional safety certification with at least 3 years of dedicated construction safety related experience. The SSHO shall ensure that the requirements of the VA and of 29 CFR 1926.16 are met for the project. The SSHO must be at the work site at all times, during construction activities, to implement and administer the Contractor's safety program and government-accepted Accident Prevention Plan. If the SSHO is off-site for a period longer than 8 hours, or one shift, and not exceeding 40 hours, or 5 shifts, a pre-approved Alternate Safety Officer shall be provided and shall fulfill the same roles and responsibilities as the primary SSHO. The Alternate Safety Officer shall have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated sub-paragraphs. When the SSHO is temporarily (not to exceed 8 hours) off-site, a Designated Representative (DR) from the Prime Contractors' staff, as identified in the AHA may be used in lieu of an Alternate Safety Officer and shall be on the project site, at all times, when work is being performed. Note that the DR is a collateral duty safety position, with safety duties in addition to their full-time occupation. Each subcontractor shall designate a minimum of one CP in compliance with 29 CFR 1926.20 (b) (2) that will be identified as a CP to administer their individual safety programs.

- B. Further, all specialized Competent Persons for the work crews will be supplied by the respective contractor as

required by 29 CFR 1926 (i.e. Asbestos, Electrical, Cranes, & Derricks, Demolition, Fall Protection, Fire Safety/Life Safety, Ladder, Rigging, Scaffolds, and Trenches/Excavations).

- C. These Competent Persons can have collateral duties as the subcontractor's superintendent and/or work crew lead persons as well as fill more than one specialized CP role (i.e. Asbestos, Electrical, Cranes, & Derricks, Demolition, Fall Protection, Fire Safety/Life Safety, Ladder, Rigging, Scaffolds, and Trenches/Excavations). However, the SSHO has be a separate qualified individual from the Prime Contractor's Superintendent and/or Quality Control Manager with duties only as the SSHO
- D. The SSHO or an equally qualified Designated Representative/alternate will maintain a presence on the site during construction operations in accordance with FAR Clause 52.236-6: *Superintendence by the Contractor*. CPs will maintain presence during their construction activities in accordance with above mentioned clause. A listing of the designated SSHO and all known CPs shall be submitted prior to the start of work as part of the APP with the training documentation and/or AHA as listed in Section 1.8 below.
- E. The repeated presence of uncontrolled hazards during a contractor's work operations will result in the designated CP as being deemed incompetent and result in the required removal of the employee in accordance with FAR Clause 52.236-5: Material and Workmanship, Paragraph (c).

**TRAINING:**

- A. The designated Prime Contractor SSHO must meet the requirements of all applicable OSHA standards and be capable (through training, experience, and qualifications) of ensuring that the requirements of 29 CFR 1926.16 and other appropriate Federal, State and local requirements are met for the project.
- B. As a minimum the SSHO must have completed the OSHA 30-hour Construction Safety Outreach class within the past five (5) years and:
  - 1. Seven (7) years of construction industry safety related experience.
  - 2. OR have a safety and health degree from an accredited university or college and five (5) years of construction industry safety related experience
  - 3. OR hold as current, a Certified Safety Professional (CSP) or a Construction Health and Safety Technician (CHST) certification and five (3) years of construction industry safety related experience.
- C. The Alternate Safety Officer shall meet, at a minimum, the requirements of EM 385-1-1 Section 1 and have five (5) years of construction industry safety experience. If the SSHO does not have a current certification, certification must be obtained within 90 days, maximum, of contract award.
- D. All designated CPs shall have completed the OSHA 30-hour Construction Safety course and/or EM 385-1-1 40-hour training within the past 5 years. In addition, all CPs with high hazard work operations (such as operations involving

asbestos, electrical, cranes, demolition, work at heights/fall protection, fire safety/life safety, ladder, rigging, scaffolds, and trenches/excavations) shall have a specialized formal course in the hazard recognition and control associated with those high hazard work operations. Documented "repeat" deficiencies in the execution of safety requirements will require retaking the requisite formal course.

- E. All other construction workers shall have the OSHA 10-hour Construction Safety Outreach course within the past 5 years and any necessary safety training to be able to identify hazards within their work environment.
- F. Submit training records associated with the above training requirements to Contracting Officer Representative for review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES 15 calendar days prior to the date of the preconstruction safety conference for acceptance.
- G. Prior to any worker for the contractor or subcontractors beginning work, they shall undergo a safety briefing provided by the SSHO or his/her designated representative. As a minimum, this briefing shall include information on the site-specific hazards, construction limits, VAMC safety guidelines, means of egress, break areas, work hours, locations of restrooms, use of VAMC equipment, emergency procedures, accident reporting etc... Documentation shall be provided to the COR that individuals have undergone contractor's safety briefing.
- H. Ongoing safety training will be accomplished in the form of weekly documented safety meeting.

## **INSPECTIONS:**

- A. The SSHO shall conduct frequent and regular safety inspections (daily) of the site and each of the subcontractors CPs shall conduct frequent and regular safety inspections (daily) of their work operations as required by 29 CFR 1926.20(b)(2). Each week, the SSHO shall conduct a formal documented inspection with a written report of the entire construction areas with the subcontractors' "Trade Safety and Health CPs" present in their work areas. Coordinate with, and report findings and corrective actions weekly to Contracting Officer Representative.
- B. A Certified Safety Professional (CSP) with specialized knowledge in construction safety or a certified Construction Safety and Health Technician (CSHT) shall randomly conduct a monthly site safety inspection. The CSP or CSHT shall be one that is not a part of the immediate site project team. The individual can be a corporate safety professional or independently contracted who is not an immediate member of the construction project site team. The CSP or CSHT will provide their certificate number on the required report for verification as necessary.
  1. Results of the inspection will be documented with tracking of the identified hazards to abatement.
  2. The Contracting Officer Representative will be notified immediately prior to start of the inspection and invited to accompany the inspection.
  3. Identified hazard and controls will be discussed to come to a mutual understanding to ensure abatement and prevent future reoccurrence.

4. A report of the inspection findings with status of abatement will be provided to the Contracting Officer Representative within one week of the onsite inspection.

**MISHAPS, OSHA 300 LOGS, AND EXPOSURE-HOURS:**

- A. The prime contractor shall establish and maintain a Near-Miss and Mishap reporting, recordkeeping, and analysis system to track and analyze all injuries and illnesses, high visibility incidents, and accidental property damage (both government and contractor) that occur on site. Notify and provide an initial report to the Contracting Officer Representative as soon as practical, but no more than two hours after any Moderate or Major Mishap, High Visibility Incidents, or any weight handling and hoisting equipment mishap. No Impact/Near-Miss and Minor Mishaps shall be reported within 24 hours or as soon as practical. Within the notification the sender shall include contractor name; contract title; type of contract; name of activity, installation or location where mishap occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of mishap (to include type of construction equipment used, PPE used, etc.). Preserve the conditions and evidence on the Mishap site until the Contracting Officer Representative determines whether a government investigation will be conducted.
- B. Conduct a mishap investigation for all Mishaps including mishaps resulting in at least \$20,000 in damages without injury, to establish the root cause(s) of the mishap. The Mishap investigation shall include images, 5 whys, the



injured person's firsthand account, any witness accounts, methods of procedures, related AHA to the task, and corrective action plan signed by the president, vice president, or appropriate corporate-level leadership identified in the company's org chart for the project. Additionally, complete the VA Form 2162 (or equivalent) and provide the report to the Contracting Officer Representative within 7 calendar days of the accident. The Contracting Officer will provide copies of any required or special forms.

- C. A summation of all exposure-hours worked by the contractor and associated sub-contractors for each month will be reported to Contracting Officer Representative no later than the 10<sup>th</sup> day of the following month.
- D. A summation of all Minor, Moderate, and Major incidents experienced on site by the contractor and associated sub-contractors for each month will be provided to the Contracting Officer Representative monthly. The contractor and associated sub-contractors' OSHA 300 logs will be made available to the Contracting Officer Representative upon request.

**PERSONAL PROTECTIVE EQUIPMENT (PPE) :**

- A. PPE is governed in all areas by the nature of the work the employee is performing. For example, specific PPE required for performing work on electrical equipment is identified in NFPA 70E, Standard for Electrical Safety in the Workplace.
- B. Any PPE above and beyond the Mandatory PPE identified below and determined by the Contracting Officer or their Representative to be necessary in the performance of the

Governments duties relative to this contract, shall be provided to the Government upon written notification by the Contracting Officer and at no additional expense to the Government. All PPE provided to the Government shall also be accompanied by any relevant or required training necessary to ensure its proper use.

C. Mandatory Minimum PPE includes:

1. Hard Hats - unless written authorization is given by the Contracting Officer Representative in circumstances of work operations that have limited potential for falling object hazards such as during finishing work or minor remodeling. With authorization to relax the requirement of hard hats, if a worker becomes exposed to an overhead falling object hazard, then hard hats would be required in accordance with the OSHA and/or EM 385-1-1 regulations.
2. Safety glasses - unless written authorization is given by the Contracting Officer Representative in circumstances of no eye hazards, appropriate safety glasses meeting the ANSI Z.87.1 standard must be worn by each person on site.
3. Appropriate Safety Shoes - based on the hazards present, safety shoes meeting the requirements of ASTM F2413-11 shall be worn by each person on site unless written authorization is given by Contracting Officer Representative in circumstances of no foot hazards.
4. Hearing protection - Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks.

5. High-visibility Safety Apparel - appropriate brightly colored clothing that feature some type of reflective material on them, such as safety vests, shall be worn by personnel or visitors on construction sites.

#### **PRE-CONSTRUCTION RISK ASSESSMENT**

- A. Control of all construction-associated hazards that affect VA medical facilities, their occupants, services and mission-essential functions and capabilities is critical in all medical center facilities. VHA Pre-Construction Risk Assessments (PCRAs) for construction, renovation and maintenance projects are included with this contract solicitation with required mitigations of identified hazards. VHA-PCRAs will be re-validated and updated as needed based on but not limited to changes from original designs, affected individuals, areas/locations, scope, contractor means and methods, safety requirements, phasing, contractor competencies and capabilities.
- B. Infection Prevention and Control is critical in all medical center facilities. Interior construction activities causing disturbance of existing dust, or creating new dust, must be conducted within ventilation-controlled areas that minimize the flow of airborne particles into patient areas. A detailed analysis of potential risks for infectious disease transmission affecting the care, treatment or services of patients or residents has been conducted. VHA Infection Control Risk Assessments (ICRAs) are included with this contract solicitation with required mitigation actions/activities. VHA-ICRAs will be re-validated and updated as needed based on changes in original designs, affected individuals, area(s) or location(s), scope, contractor means and methods, infection prevention and

control requirements, differing site conditions, phasing, contractor competencies and capabilities, and infectious disease outbreaks.

C. For work occurring at a VA medical facility, coordinate with the facility Safety Manager/Officer, as several aspects of this section directly relate to infection control risk assessments required in or adjacent to construction affecting occupied buildings accredited by The Joint Commission.

D. Products and Materials:

1. Sheet Plastic: Fire retardant polyethylene, 6-mil thickness meeting local fire codes
2. Barrier Doors: Self Closing // One-hour // Two-hour // fire-rated // solid core wood in steel frame, painted
3. Dust proof // one-hour // two-hour // fire-rated // drywall
4. High Efficiency Particulate Air-Equipped filtration machine rated at 95% capture of 0.3 microns including pollen, mold spores and dust particles. HEPA filters should have ASHRAE 85 or other prefilter to extend the useful life of the HEPA. Provide both primary and secondary filtrations units. Maintenance of equipment and replacement of the HEPA filters and other filters will be in accordance with manufacturer's instructions.
5. Exhaust Hoses: Heavy duty, flexible steel reinforced; Ventilation Blower Hose
6. Adhesive Walk-off Mats: Provide minimum size mats of 24 inches x 36 inches

7. Disinfectant: EPA-registered, Hospital-approved disinfectant or equivalent product

8. Portable Ceiling Access Module

#### **DUST CONTROL**

- A. Contractor shall verify that dust will not be introduced into the facility through intake vents or building openings. HEPA filtration on all intake vents is required where dust may be introduced.
- B. Dust created from disturbance of soil such as from vehicle movement will be wetted with use of a water truck as necessary.
- C. All cutting, drilling, grinding, sanding, or disturbance of materials shall be accomplished with tools equipped with either local exhaust ventilation (i.e. vacuum systems) or wet suppression controls.

#### **FIRE SAFETY**

- A. Fire Safety Plan: Establish and maintain a site-specific fire protection program in accordance with 29 CFR 1926. Prior to start of work, prepare a plan detailing project-specific fire safety measures, including periodic status reports, and submit to Contracting Officer Representative review for compliance with contract requirements in accordance with Section 01 33 23, SHOP DRAWINGS, PRODUCT DATA AND SAMPLES. This plan may be an element of the Accident Prevention Plan.
- B. Site and Building Access: Maintain free and unobstructed access to facility emergency services and for fire, police

and other emergency response forces in accordance with NFPA 241.

- C. For work occurring at a VA medical facility, coordinate with the facility Safety Manager/Officer, as several aspects of this section directly relate to interim life safety measures required in or adjacent to construction affecting occupied buildings accredited by The Joint Commission.
- D. Separate temporary facilities, such as trailers, storage sheds, and dumpsters, from existing buildings and new construction by distances in accordance with NFPA 241. For small facilities with less than 6 m (20 feet) exposing overall length, separate by 3m (10 feet).
- E. Temporary Construction Partitions:
  - 1. Install and maintain temporary construction partitions to provide smoke-tight separations between // construction areas // the areas that are described in phasing requirements // and adjoining areas. Construct partitions of gypsum board or fire-retardant treated plywood (fire-retardant treated in accordance with NFPA 703) on both sides of fire-retardant treated wood framing or metal steel studs. Extend the partitions through suspended ceilings to floor slab deck or roof. Seal joints and penetrations. At door openings, install Class C,  $\frac{3}{4}$  hour fire/smoke rated doors with self-closing devices.
  - 2. Install // one-hour // two-hour // fire-rated // temporary construction partitions as shown on drawings to maintain integrity of existing exit stair enclosures, exit passageways, fire-rated enclosures of hazardous

areas, horizontal exits, smoke barriers, vertical shafts and openings enclosures.

3. Close openings in smoke barriers and fire-rated construction to maintain fire ratings. Seal penetrations with listed through-penetration firestop materials in accordance with Section 07 84 00, FIRESTOPPING.

F. Temporary Heating and Electrical: Install, use and maintain installations in accordance with 29 CFR 1926, NFPA 241 and NFPA 70.

G. Means of Egress: Do not block exiting for occupied buildings, including paths from exits to roads. Minimize disruptions and coordinate with Contracting Officer Representative.

H. Egress Routes for Construction Workers: Maintain free and unobstructed egress. Inspect daily. Report findings and corrective actions weekly to Contracting Officer Representative.

I. Fire Extinguishers: Provide and maintain extinguishers in construction areas and temporary storage areas in accordance with 29 CFR 1926, NFPA 241 and NFPA 10.

J. Flammable and Combustible Liquids: Store, dispense and use liquids in accordance with 29 CFR 1926, NFPA 241 and NFPA 30.

K. Standpipes: Install and extend standpipes up with each floor in accordance with 29 CFR 1926 and NFPA 241. Do not charge wet standpipes subject to freezing until weather protected.

- L. Sprinklers: Install, test and activate new automatic sprinklers prior to removing existing sprinklers.
- M. Existing Fire Protection: Do not impair automatic sprinklers, smoke and heat detection, and fire alarm systems, except for portions immediately under construction, and temporarily for connections. Provide fire watch for impairments more than 4 hours in a 24-hour period. Request interruptions in accordance with Article, OPERATIONS AND STORAGE AREAS, and coordinate with Contracting Officer Representative . All existing or temporary fire protection systems (fire alarms, sprinklers) located in construction areas shall be tested as coordinated with the medical center. Parameters for the testing and results of any tests performed shall be recorded by the medical center and copies provided to the Resident Engineer.
- N. Smoke Detectors: Prevent accidental operation. Remove temporary covers at end of work operations each day. Coordinate with Contracting Officer Representative.
- O. Hot Work: Perform and safeguard hot work operations in accordance with NFPA 241 and NFPA 51B. Coordinate with Facility Engineer / Facility Safety Office. at least \_\_24\_\_ hours in advance.
- P. Fire Hazard Prevention and Safety Inspections: Inspect entire construction areas weekly. Coordinate with, and report findings and corrective actions weekly to Contracting Officer Representative.
- Q. Smoking: Smoking is prohibited in and adjacent to construction areas inside existing buildings and additions under construction. In separate and detached buildings



under construction, smoking is prohibited except in designated smoking rest areas.

R. Dispose of waste and debris in accordance with NFPA 241. Remove from buildings daily.

S. If required, submit documentation to the COR that personnel have been trained in the fire safety aspects of working in areas with impaired structural or compartmentalization features.

## **ELECTRICAL**

A. All electrical work shall comply with NFPA 70 (NEC), NFPA 70B, NFPA 70E, 29 CFR Part 1910 Subpart J - General Environmental Controls, 29 CFR Part 1910 Subpart S - Electrical, and 29 CFR 1926 Subpart K in addition to other references required by contract.

B. All qualified persons performing electrical work under this contract shall be licensed journeyman or master electricians. All apprentice electricians performing under this contract shall be deemed unqualified persons unless they are working under the immediate supervision of a licensed electrician or master electrician.

C. All electrical work will be accomplished de-energized and in the Electrically Safe Work Condition (refer to NFPA 70E for Work Involving Electrical Hazards, including Exemptions to Work Permit). Any Contractor, subcontractor or temporary worker who fails to fully comply with this requirement is subject to immediate termination in accordance with FAR clause 52.236-5(c). Only in rare circumstance when achieving an electrically safe work condition prior to beginning work would increase or cause additional hazards or is infeasible due to equipment design

or operational limitations is energized work permitted. The Contracting Officer Representative with approval of the Medical Center Director will make the determination if the circumstances would meet the exception outlined above. An AHA and permit specific to energized work activities will be developed, reviewed, and accepted by the VA prior to the start of that activity.

1. Development of a Hazardous Electrical Energy Control Procedure is required prior to de-energization. A single Simple Lockout/Tagout Procedure for multiple work operations can only be used for work involving qualified person(s) de-energizing one set of conductors or circuit part source. Task specific Complex Lockout/Tagout Procedures are required at all other times.

2. Verification of the absence of voltage after de-energization and lockout/tagout is considered "energized electrical work" (live work) under NFPA 70E, and shall only be performed by qualified persons wearing appropriate shock protective (voltage rated) gloves and arc rate personal protective clothing and equipment, using Underwriters Laboratories (UL) tested and appropriately rated contact electrical testing instruments or equipment appropriate for the environment in which they will be used.

3. Personal Protective Equipment (PPE) and electrical testing instruments will be readily available for inspection by the Contracting Officer Representative.

D. Before beginning any electrical work, an Activity Hazard Analysis (AHA) will be conducted to include Shock Hazard and Arc Flash Hazard analyses (NFPA Tables can be used only

as a last alternative and it is strongly suggested a full Arc Flash Hazard Analyses be conducted). Work shall not begin until the AHA for the work activity and permit for energized work has been reviewed and accepted by Contracting Officer Representative and discussed with all engaged in the activity, including the Contractor, subcontractor(s), and Government on-site representatives at preparatory and initial control phase meetings.

- E. Ground-fault circuit interrupters. GFCI protection shall be provided where an employee is operating or using cord- and plug-connected tools related to construction activity supplied by 125-volt, 15-, 20-, or 30- ampere circuits. Where employees operate or use equipment supplied by greater than 125-volt, 15-, 20-, or 30- ampere circuits, GFCI protection or an assured equipment grounding conductor program shall be implemented in accordance with NFPA 70E - 2015, Chapter 1, Article 110.4(C)(2).

#### **FALL PROTECTION**

- A. The fall protection (FP) threshold height requirement is 4ft for ALL WORK in government occupied sites, unless the OSHA 29 CFR 1926 or EM 385-1-1 requirements are more stringent, to include steel erection activities, systems-engineered activities (prefabricated) metal buildings, residential (wood) construction and scaffolding work. For work occurring at new "greenfield" sites or sites with 24/7 Contractor controlled 100% restricted access, the fall protection (FP) threshold height may be increased to 6ft unless the OSHA 29 CFR 1926 or EM 385-1-1 requirements are more stringent.

1. The use of a Safety Monitoring System (SMS) as a fall protection method is prohibited.
2. The use of Controlled Access Zone (CAZ) as a fall protection method is prohibited.
3. A Warning Line System (WLS) may ONLY be used on floors or flat or low-sloped roofs (between 0 - 18.4 degrees or 4:12 slope) and shall be erected around all sides of the work area (See 29 CFR 1926.502(f) for construction of WLS requirements). Working within the WLS does not require FP. No worker shall be allowed in the area between the roof or floor edge and the WLS without FP. FP is required when working outside the WLS.
4. Fall protection while using a ladder will be governed by the more stringent of OSHA and EM 385-1-1 requirements.

#### **CRANES**

- A. All crane work shall comply with 29 CFR 1926 Subpart CC and EM 385-1-1.
- B. Prior to operating a crane, the operator must be licensed, qualified or certified to operate the crane. Thus, all the provisions contained with Subpart CC are effective and there is no "Phase In" date.
- C. A detailed lift plan for all lifts shall be submitted to the Resident Engineer, COR and/or other Government Designated Authority 14 days prior to the scheduled lift complete with route for truck carrying load, crane load analysis, siting of crane and path of swing and all other elements of a critical lift plan where the lift meets the definition of a critical lift. Critical lifts require a more comprehensive lift plan to minimize the potential of

crane failure and/or catastrophic loss. The plan must be reviewed and accepted by the General Contractor before being submitted to the VA for review. The lift will not be allowed to proceed without prior acceptance of this document by the COR and/or Facility Safety.

D. Crane operators shall not carry loads

1. over the general public or VAMC personnel
2. over any occupied building unless
  - a. The top two or more floors are vacated under the lift affected area AND a Contractor provided engineering analysis, performed by a Professional Engineer, demonstrates that overhead protection provided by the roof and vacated floors is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater
  - b. OR Overhead protection is designed by a Professional Engineer and installed over the roof by the Contractor to protect the occupants below. The installed system shall be sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.
  - c. OR A formal request may be submitted to the Contracting Officer for documentation, if available, indicating the roof of the building has been previously designed and constructed using a reinforced concrete roof slab or other means to provide overhead protection from construction activities specifically associated with the vertical or horizontal expansion

of the same building AND a Contractor provided engineering analysis, performed by a Professional Engineer, demonstrates that overhead protection provided by the roof is sufficient to protect against the potential loss of a load being lifted or a minimum of 300 psf of protection is provided, whichever is greater.

#### **CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)**

- A. All installation, maintenance, and servicing of equipment or machinery shall comply with 29 CFR 1910.147 except for specifically referenced operations in 29 CFR 1926 such as concrete & masonry equipment [1926.702(j)], heavy machinery & equipment [1926.600(a)(3)(i)], and process safety management of highly hazardous chemicals (1926.64). Control of hazardous electrical energy during the installation, maintenance, or servicing of electrical equipment shall comply with Section 1.15 to include NFPA 70E and other VA specific requirements discussed in the section.

#### **CONFINED SPACE ENTRY**

- A. All confined space entry shall comply with 29 CFR 1926, Subpart AA except for specifically referenced operations in 29 CFR 1926 such as excavations/trenches [1926.651(g)].
- B. A site-specific Confined Space Entry Plan (including permitting process) shall be developed and submitted to the Facility Engineer, COR and/or Facility Safety.

#### **WELDING AND CUTTING**

As specified in section, Hot Work: Perform and safeguard hot work operations in accordance with NFPA 241 and NFPA

51B. Coordinate with COR. Obtain permits from Facility Engineer and or COR at least \_\_24\_\_ hours in advance.

## **1.26 LADDERS**

- A. All Ladder use shall comply with 29 CFR 1926 Subpart X and EM 385-1-1.
- B. All portable ladders shall be of sufficient length and shall be placed so that workers will not stretch or assume a hazardous position.
- C. Manufacturer safety labels shall be in place on ladders
- D. Step Ladders shall not be used in the closed position
- E. Top steps or cap of step ladders shall not be used as a step
- F. Portable ladders, used as temporary access, shall extend at least 3 ft (0.9 m) above the upper landing surface.
  - 1. When a 3 ft (0.9-m) extension is not possible, a grasping device (such as a grab rail) shall be provided to assist workers in mounting and dismounting the ladder.
  - 2. In no case shall the length of the ladder be such that ladder deflection under a load would, by itself, cause the ladder to slip from its support.
- G. Ladders shall be inspected for visible defects on a daily basis and after any occurrence that could affect their safe use. Broken or damaged ladders shall be immediately tagged "DO NOT USE," or with similar wording, and withdrawn from service until restored to a condition meeting their original design.

## **FLOOR & WALL OPENINGS**

- A. All floor and wall openings shall comply with 29 CFR 1926 Subpart M and EM 385-1-1.
- B. Floor and roof holes/openings are any that measure over 2 in (51 mm) in any direction of a walking/working surface which persons may trip or fall into or where objects may fall to the level below. Skylights located in floors or roofs are considered floor or roof hole/openings.
- C. All floor, roof openings or hole into which a person can accidentally walk or fall through shall be guarded either by a railing system with toe boards along all exposed sides or a load-bearing cover. When the cover is not in place, the opening or hole shall be protected by a removable guardrail system or shall be attended when the guarding system has been removed, or other fall protection system.
  - 1. Covers shall be capable of supporting, without failure, at least twice the weight of the worker, equipment and material combined.
  - 2. Covers shall be secured when installed, clearly marked with the word "HOLE", "COVER" or "Danger, Roof Opening-Do Not Remove" or color-coded or equivalent methods (e.g., red or orange "X"). Workers must be made aware of the meaning for color coding and equivalent methods.
  - 3. Roofing material, such as roofing membrane, insulation or felts, covering or partly covering openings or holes, shall be immediately cut out. No hole or opening shall be left unattended unless covered.



4. Non-load-bearing skylights shall be guarded by a load-bearing skylight screen, cover, or railing system along all exposed sides.
5. Workers are prohibited from standing/walking on skylights.

DRAFT

# APPENDIX B

## GENERAL REQUIREMENTS

### GENERAL

#### SAFETY REQUIREMENTS

Refer to section 01 35 26, SAFETY REQUIREMENTS for safety and infection control requirements.

#### GENERAL INTENTION

Contractor shall completely prepare site for building operations, including demolition and removal of existing structures, and furnish labor and materials and perform work for deinstallation of steam cart washer, site preparation, and installation of Electric Cart Washer as required by drawings and specifications.

Visits to the site by Bidders may be made only by appointment with the Contracting Officer Representative, Jaime Noyola.

Before placement and installation of work subject to tests by testing laboratory retained by Department of Veterans Affairs, the Contractor shall notify the COR in sufficient time to enable testing laboratory personnel to be present at the site in time for proper taking and testing of specimens and field inspection. Such prior notice shall be not less than three workdays unless otherwise designated by the COR.

All employees of general contractor and subcontractors shall comply with VA security management program and obtain

permission of the VA police, be identified by project and employer, and restricted from unauthorized access.

#### **STATEMENT OF BID ITEM(S)**

ITEM I, GENERAL CONSTRUCTION: Work includes general construction, electrical alterations, utility/critical water, drainage, necessary removal of existing structures and construction and certain other items.

#### **SPECIFICATIONS AND DRAWINGS FOR CONTRACTOR**

Drawings and contract documents may be obtained from the website where the solicitation is posted. Additional copies will be at Contractor's expense.

#### **CONSTRUCTION SECURITY REQUIREMENTS**

Security Plan:

The security plan defines both physical and administrative security procedures that will remain effective for the entire duration of the project.

The General Contractor is responsible for assuring that all sub-contractors working on the project and their employees also comply with these regulations.

Security Procedures:

General Contractor's employees shall not enter the project site without appropriate badge. They may also be subject to inspection of their personal effects when entering or leaving the project site.

Before starting work the General Contractor shall give one week's notice to the Contracting Officer so that security escort can be provided for the employees. This notice is separate from any notices required for utility shutdown described later in this section.

No photography of VA premises is allowed without written permission of the Contracting Officer. Patients and staff are not to be photographed at any time.

VA reserves the right to close down or shut down the project site and order General Contractor's employees off the premises in the event of a national emergency. The General Contractor may return to the site only with the written approval of the Contracting Officer.

#### Motor Vehicle Restrictions

Vehicle authorization request shall be required for any vehicle entering the site and such request shall be submitted 24 hours before the date and time of access. Access shall be restricted to picking up and dropping off materials and supplies.

An unlimited number of permits shall be issued for General Contractor and its employees for parking in designated areas only. Contractor to coordinate with COR.

#### **OPERATIONS AND STORAGE AREAS (FAR 52.236-10)**

The Contractor shall confine all operations (including storage of materials) on Government premises to areas authorized or approved by the Contracting Officer. The Contractor shall hold and save the Government, its officers and agents, free and harmless from liability of any nature occasioned by the Contractor's performance.

Temporary buildings (e.g., storage sheds, shops, offices) and utilities may be erected by the Contractor only with the approval of the Contracting Officer and shall be built with labor and materials furnished by the Contractor without expense to the Government. The temporary buildings and utilities shall remain the property of the Contractor and

shall be removed by the Contractor at its expense upon completion of the work. With the written consent of the Contracting Officer, the buildings and utilities may be abandoned and need not be removed.

The Contractor shall, under regulations prescribed by the Contracting Officer, use only established roadways, or use temporary roadways constructed by the Contractor when and as authorized by the Contracting Officer. When materials are transported in prosecuting the work, vehicles shall not be loaded beyond the loading capacity recommended by the manufacturer of the vehicle or prescribed by any Federal, State, or local law or regulation. When it is necessary to cross curbs or sidewalks, the Contractor shall protect them from damage. The Contractor shall repair or pay for the repair of any damaged curbs, sidewalks, or roads.

Working space and space available for storing materials shall be as determined by the COR.

Workers are subject to rules of Medical Center applicable to their conduct.

Execute work in such a manner as to interfere as little as possible with work being done by others. Keep roads clear of construction materials, debris, standing construction equipment and vehicles at all times.

Execute work so as to interfere as little as possible with normal functioning of Medical Center as a whole, including operations of utility services, fire protection systems and any existing equipment, and with work being done by others. Use of equipment and tools that transmit vibrations and noises through the building structure, are not permitted in buildings that are occupied, during construction, jointly by patients or medical personnel, and Contractor's

personnel, except as permitted by COR where required by limited working space.

Do not store materials and equipment in other than assigned areas.

Schedule delivery of materials and equipment to immediate construction working areas within buildings in use by Department of Veterans Affairs in quantities sufficient for not more than two work days. Provide unobstructed access to Medical Center areas required to remain in operation.

Where access by Medical Center personnel to vacated portions of buildings is not required, storage of Contractor's materials and equipment will be permitted subject to fire and safety requirements.

Utilities Services: Where necessary to cut existing pipes, electrical wires, conduits, cables, etc., of utility services, or of fire protection systems or communications systems (except telephone), they shall be cut and capped at suitable places where shown; or, in absence of such indication, where directed by COR. All such actions shall be coordinated with the COR or Utility Company involved:

Whenever it is required that a connection fee be paid to a public utility provider for new permanent service to the construction project, for such items as water, sewer, electricity, gas or steam, payment of such fee shall be the responsibility of the Government and not the Contractor.

Phasing:

The Medical Center must maintain its operation 0800-1630, 5 days a week. Therefore, any interruption in service must be scheduled and coordinated with the COR to ensure that no lapses in operation occur. It is the CONTRACTOR'S

responsibility to develop a work plan and schedule detailing, at a minimum, the procedures to be employed, the equipment and materials to be used, the interim life safety measure to be used during the work, and a schedule defining the duration of the work with milestone subtasks. The work to be outlined shall include, but not be limited to:

To ensure such executions, Contractor shall furnish the COR with a schedule of phasing on which the Contractor intends to accomplish work in each specific area of site, building or portion thereof. In addition, Contractor shall notify the COR two weeks in advance of the proposed date of starting work in each specific area of site, building or portion thereof. Arrange such phasing to ensure accomplishment of this work in successive phases mutually agreeable to the COR and Contractor, as follows:

Phase I: Deinstallation of Steam Cart Washer

Phase II: Site preparation work for installation of Electric Cart Washer

Phase III: Installation of Electric Cart Washer.

Phase IV: Testing and Turn-key Completion.

Utilities Services: Maintain existing utility services for Medical Center at all times. Provide temporary facilities, labor, materials, equipment, connections, and utilities to assure uninterrupted services. Where necessary to cut existing water, steam, gases, sewer or air pipes, or conduits, wires, cables, etc. of utility services or of fire protection systems and communications systems (including telephone), they shall be cut and capped at

suitable places where shown; or, in absence of such indication, where directed by COR.

No utility service such as water, gas, steam, sewers or electricity, or fire protection systems and communications systems may be interrupted without prior approval of COR. Electrical work shall be accomplished with all affected circuits or equipment de-energized. When an electrical outage cannot be accomplished, work on any energized circuits or equipment shall not commence without a detailed work plan, the Medical Center Director's prior knowledge and written approval. Refer to specification Sections 26 05 11, REQUIREMENTS FOR ELECTRICAL INSTALLATIONS, 27 05 11 REQUIREMENTS FOR COMMUNICATIONS INSTALLATIONS and 28 05 00, COMMON WORK RESULTS FOR ELECTRONIC SAFETY AND SECURITY for additional requirements.

Contractor shall submit a request to interrupt any such services to COR, in writing, 7 days in advance of proposed interruption. Request shall state reason, date, exact time of, and approximate duration of such interruption.

Contractor will be advised (in writing) of approval of request, or of which other date and/or time such interruption will cause least inconvenience to operations of Medical Center. Interruption time approved by Medical Center may occur at other than Contractor's normal working hours.

Major interruptions of any system must be requested, in writing, at least 15 calendar days prior to the desired time and shall be performed as directed by the COR.



In case of a contract construction emergency, service will be interrupted on approval of COR. Such approval will be confirmed in writing as soon as practical.

Whenever it is required that a connection fee be paid to a public utility provider for new permanent service to the construction project, for such items as water, sewer, electricity, gas or steam, payment of such fee shall be the responsibility of the Government and not the Contractor.

Abandoned Lines: All service lines such as wires, cables, conduits, ducts, pipes and the like, and their hangers or supports, shall be removed back to their source. Those which are indicated to be abandoned but are not required to be entirely removed, shall be sealed, capped or plugged at the main, branch or panel they originate from. The lines shall not be capped in finished areas, but shall be removed and sealed, capped or plugged in ceilings, within furred spaces, in unfinished areas, or within walls or partitions; so that they are completely behind the finished surfaces.

To minimize interference of construction activities with flow of Medical Center traffic, comply with the following:

Keep roads, walks and entrances to grounds, to parking and to occupied areas of buildings clear of construction materials, debris and standing construction equipment and vehicles.

## **ALTERATIONS**

Survey: Before any work is started, the Contractor shall make a thorough survey with the COR and a representative of VA Facilities Management, of areas of building in which alterations occur and areas which are anticipated routes of

access, and furnish a report, signed by both, to the Contracting Officer. This report shall list by rooms and spaces:

Existing condition and types of resilient flooring, doors, windows, walls and other surfaces not required to be altered throughout affected areas of building.

Existence and conditions of items such as plumbing fixtures and accessories, electrical fixtures, equipment, venetian blinds, shades, etc., required by drawings to be either reused or relocated, or both.

Shall note any discrepancies between drawings and existing conditions at site.

Shall designate areas for working space, materials storage and routes of access to areas within buildings where alterations occur and which have been agreed upon by Contractor and COR.

Any items required by drawings to be either reused or relocated or both, found during this survey to be nonexistent, or in opinion of COR , to be in such condition that their use is impossible or impractical, shall be furnished and/or replaced by Contractor with new items in accordance with specifications which will be furnished by Government. Provided the contract work is changed by reason of this subparagraph B, the contract will be modified accordingly, under provisions of clause entitled "DIFFERING SITE CONDITIONS" (FAR 52.236-2) and "CHANGES" (FAR 52.243-

Re-Survey: Thirty days before expected partial or final inspection date, the Contractor and COR together shall make a thorough re-survey of the areas of buildings involved. They shall furnish a report on conditions then existing, of resilient flooring, doors, windows, walls and other

surfaces as compared with conditions of same as noted in first condition survey report:

Re-survey report shall also list any damage caused by Contractor to such flooring and other surfaces, despite protection measures; and, will form basis for determining extent of repair work required of Contractor to restore damage caused by Contractor's workers in executing work of this contract.

Protection: Provide the following protective measures:

Wherever existing roof surfaces are disturbed they shall be protected against water infiltration. In case of leaks, they shall be repaired immediately upon discovery.

Temporary protection against damage for portions of existing structures and grounds where work is to be done, materials handled and equipment moved and/or relocated.

Protection of interior of existing structures at all times, from damage, dust and weather inclemency. Wherever work is performed, floor surfaces that are to remain in place shall be adequately protected prior to starting work, and this protection shall be maintained intact until all work in the area is completed.

#### **DISPOSAL AND RETENTION**

Materials and equipment accruing from work removed and from demolition of buildings or structures, or parts thereof, shall be disposed of as follows:

Reserved items which are to remain property of the Government are identified by attached tags or noted on drawings or in specifications as items to be stored. Items that remain property of the Government shall be removed or dislodged from present locations in such a manner as to prevent damage which would be detrimental to

re-installation and reuse. Store such items where directed by COR.

Items not reserved shall become property of the Contractor and be removed by Contractor from Medical Center.

Items of portable equipment and furnishings located in rooms and spaces in which work is to be done under this contract shall remain the property of the Government. When rooms and spaces are vacated by the Department of Veterans Affairs during the alteration period, such items which are NOT required by drawings and specifications to be either relocated or reused will be removed by the Government in advance of work to avoid interfering with Contractor's operation.

**PROTECTION OF EXISTING VEGETATION, STRUCTURES, EQUIPMENT, UTILITIES, AND IMPROVEMENTS (FAR 52.236-9)**

The Contractor shall preserve and protect all structures, equipment, and vegetation (such as trees, shrubs, and grass) on or adjacent to the work site, which are not to be removed and which do not unreasonably interfere with the work required under this contract. The Contractor shall only remove trees when specifically authorized to do so, and shall avoid damaging vegetation that will remain in place. If any limbs or branches of trees are broken during contract performance, or by the careless operation of equipment, or by workers, the Contractor shall trim those limbs or branches with a clean cut and paint the cut with a tree-pruning compound as directed by the Contracting Officer.

The Contractor shall protect from damage all existing improvements and utilities at or near the work site and on

adjacent property of a third party, the locations of which are made known to or should be known by the Contractor. The Contractor shall repair any damage to those facilities, including those that are the property of a third party, resulting from failure to comply with the requirements of this contract or failure to exercise reasonable care in performing the work. If the Contractor fails or refuses to repair the damage promptly, the Contracting Officer may have the necessary work performed and charge the cost to the Contractor.

Refer to Section 01 57 19, TEMPORARY ENVIRONMENTAL CONTROLS, for additional requirements on protecting vegetation, soils and the environment. Refer to Articles, "Alterations", "Restoration", and "Operations and Storage Areas" for additional instructions concerning repair of damage to structures and site improvements.

Refer to FAR clause 52.236-7, "Permits and Responsibilities," which is included in General Conditions. A National Pollutant Discharge Elimination System (NPDES) permit is required for this project. The Contractor is considered an "operator" under the permit and has extensive responsibility for compliance with permit requirements. VA will make the permit application available at the (appropriate medical center) office. The apparent low bidder, contractor and affected subcontractors shall furnish all information and certifications that are required to comply with the permit process and permit requirements. Many of the permit requirements will be satisfied by completing construction as shown and specified. Some requirements involve the Contractor's method of operations and operations planning and the

Contractor is responsible for employing best management practices. The affected activities often include, but are not limited to the following:

Designating areas for equipment maintenance and repair;  
Providing waste receptacles at convenient locations and provide regular collection of wastes;

Locating equipment wash down areas on site, and provide appropriate control of wash-waters;

Providing protected storage areas for chemicals, paints, solvents, fertilizers, and other potentially toxic materials; and

Providing adequately maintained sanitary facilities.

#### **RESTORATION**

Remove, cut, alter, replace, patch and repair existing work as necessary to install new work. Except as otherwise shown or specified, do not cut, alter or remove any structural work, and do not disturb any ducts, plumbing, steam, gas, or electric work without approval of the COR. Existing work to be altered or extended and that is found to be defective in any way, shall be reported to the COR before it is disturbed. Materials and workmanship used in restoring work, shall conform in type and quality to that of original existing construction, except as otherwise shown or specified.

Upon completion of contract, deliver work complete and undamaged. Existing work (walls, ceilings, partitions, floors, mechanical and electrical work, lawns, paving, roads, walks, etc.) disturbed or removed as a result of performing required new work, shall be patched, repaired, reinstalled, or replaced with new work, and refinished and

left in as good condition as existed before commencing work.

At Contractor's own expense, Contractor shall immediately restore to service and repair any damage caused by Contractor's workers to existing piping and conduits, wires, cables, etc., of utility services or of fire protection systems and communications systems (including telephone) which are not scheduled for discontinuance or abandonment.

Expense of repairs to such utilities and systems not shown on drawings or locations of which are unknown will be covered by adjustment to contract time and price in accordance with clause entitled "CHANGES" (FAR 52.243-4) and "DIFFERING SITE CONDITIONS" (FAR 52.236-2).

#### **PHYSICAL DATA - SOIL CONDITIONS**

Data and information furnished or referred to below is for the Contractor's information. The Government shall not be responsible for any interpretation of or conclusion drawn from the data or information by the Contractor.

The indications of physical conditions on the drawings and in the specifications are the result of site investigations by Contractor.

#### **LAYOUT OF WORK**

The Contractor shall lay out the work from Government established base lines and bench marks, indicated on the drawings, and shall be responsible for all measurements in connection with the layout. The Contractor shall furnish, at Contractor's own expense, all stakes, templates, platforms, equipment, tools, materials, and labor required

to lay out any part of the work. The Contractor shall be responsible for executing the work to the lines and grades that may be established or indicated by the Contracting Officer. The Contractor shall also be responsible for maintaining and preserving all stakes and other marks established by the Contracting Officer until authorized to remove them. If such marks are destroyed by the Contractor or through Contractor's negligence before their removal is authorized, the Contracting Officer may replace them and deduct the expense of the replacement from any amounts due or to become due to the Contractor.

**(FAR 52.236-17)**

#### **AS-BUILT DRAWINGS**

The contractor shall maintain two full size sets of as-built drawings which will be kept current during construction of the project, to include all contract changes, modifications and clarifications.

All variations shall be shown in the same general detail as used in the contract drawings. To ensure compliance, as-built drawings shall be made available for the COR review, as often as requested.

Contractor shall deliver two approved completed sets of as-built drawings in the electronic version (scanned PDF) to the COR within 15 calendar days after each completed phase and after the acceptance of the project by the COR.

Paragraphs A, B, & C shall also apply to all shop drawings.

#### **WARRANTY MANAGEMENT**

Warranty Management Plan: Develop a warranty management plan which contains information relevant to FAR 52.246-21

Warranty of Construction in VCB HCC at least 30 days before the planned pre-warranty conference, submit one set the warranty management plan. Include within the warranty



management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan must be in narrative form and contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesman, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was approved. Warranty information made available during the construction phase must be submitted to the Contracting Officer for approval prior to each monthly invoice for payment. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period will begin on the date of the project acceptance and continue for the product warranty period. A joint 4 month and 9 month warranty inspection will be conducted, measured from time of acceptance, by the Contactor and the Contracting Officer. Include in the warranty management plan, but not limited to, the following:

1. Roles and responsibilities of all personnel associated with the warranty process, including points of contact and telephone numbers within the company of the Contractor, subcontractors, manufacturers or suppliers involved.
2. Furnish with each warranty the name, address and telephone number of each of the guarantor's representatives nearest project location.
3. Listing and status of delivery of all Certificates of Warranty for extended warranty items, to include roofs, HVAC balancing, pumps, motors, transformers and for all

commissioned systems such as fire protection and alarm systems, sprinkler systems and lightning protection systems, etc.

4. A list for each warranted equipment item, feature of construction or system indicating:
  - a. Name of item.
  - b. Model and serial numbers.
  - c. Location where installed.
  - d. Name and phone numbers of manufacturers and suppliers.
  - e. Name and phone numbers of manufacturers or suppliers.
  - f. Names, addresses and phone numbers of sources of spare parts.
  - g. Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have extended warranties must be indicated with separate warranty expiration dates.
  - h. Starting point and duration of warranty period.
  - i. Summary of maintenance procedures required to continue the warranty in force.
  - j. Cross-reference to specific pertinent Operation and Maintenance manuals.
  - k. Organizations, names and phone numbers of persons to call for warranty service.
  - l. Typical response time and repair time expected for various warranted equipment.

5. The plans for attendance at the 4 and 9-month post construction warranty inspections conducted by the government.
6. Procedure and status of tagging of all equipment covered by extended warranties.
7. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty and/or safety reasons.

Performance & Payment Bonds: The Performance & Payment Bonds must remain effective throughout the construction period.

1. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
2. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the contractor's expenses, the Contracting Officer will have the right to recoup expenses from the bonding company.
3. Following oral or written notification of required construction warranty repair work, the Contractor shall respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

Pre-Warranty Conference: Prior to contract completion, and at a time designated by the Contracting Officer, the Contractor shall meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. Communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty will be established/reviewed at this meeting. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contract will be located within the local service area of the warranted construction, be continuously available and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in conjunction with other portions of this provision.

#### Contractor's Response to Construction Warranty Service Requirements:

Following oral or written notification by the Contracting Officer, the Contractor shall respond to construction warranty service requirements in accordance with the "Construction Warranty Service Priority List" and the three categories of priorities listed below. Submit a report on any warranty item that has been repaired during the warranty period. Include within the report the cause of the problem, date reported, corrective action taken, and when

the repair was completed. If the Contractor does not perform the construction warranty within the timeframe specified, the Government will perform the work and back charge the construction warranty payment item established.

1. First Priority Code 1. Perform onsite inspection to evaluate situation, and determine course of action within 4 hours, initiate work within 6 hours and work continuously to completion or relief.
2. Second Priority Code 2. Perform onsite inspection to evaluate situation, and determine course of action within 8 hours, initiate work within 24 hours and work continuously to completion or relief.
3. Third Priority Code 3. All other work to be initiated within 3 work days and work continuously to completion or relief.
4. The "Construction Warranty Service Priority List" is as follows:

a) Code 1-Life Safety Systems

- 1) Fire suppression systems.
- 2) Fire alarm system(s).

b) Code 1-Air Conditioning Systems

- 1) Air conditioning leak in part of the building, if causing damage.
- 2) Air conditioning system not cooling properly.

c) Code 1 Doors

- 1) Overhead doors not operational, causing a security, fire or safety problem.

Interior, exterior personnel doors or hardware, not functioning properly, causing security, fire or safety problem.

d) Code 3-Doors

- 1) Overhead doors not operational.
- 2) Interior/exterior personnel doors or hardware not functioning properly.

e) Code 1-Electrical

- 1) Power failure (entire area or any building operational after 1630 hours).
- 2) Security lights.
- 3) Smoke detectors.

f) Code 2-Electrical

- 1) Power failure (no power to a room or part of building). Receptacle and lights not operational (in a room or part of building).

g) Code 3-Electrical

- 1) Exterior lights not operational.

h) Code 1-Gas

- 1) Leaks and pipeline breaks.

i) Code 1-Heat

- 1) Power failure affecting heat.

j) Code 1-Plumbing

- 1) Hot water heater failure.

Leaking water supply pipes

k) Code 2-Plumbing

- 1) Flush valves not operating properly
- 2) Fixture drain, supply line or any water pipe leaking.
- 3) Toilet leaking at base.

l) Code 3- Plumbing

- 1) Leaky faucets.

m) Code 3-Interior

- 1) Floors damaged.
- 2) Paint chipping or peeling.
- 3) Casework damaged.

n) Code 1-Roof Leaks

- 1) Damage to property is occurring.

o) Code 2-Water (Exterior)

- 1) No water to facility.

p) Code 2-Water (Hot)

- 1) No hot water in portion of building listed.

q) Code 3

- 1) All work not listed above.

Warranty Tags: At the time of installation, tag each warranted item with a durable, oil and water-resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC

signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

| Warranty Tags                        |
|--------------------------------------|
| Type of product/material             |
| Model number                         |
| Serial number                        |
| Contract number                      |
| Warranty period from/to              |
| Inspector's signature                |
| Construction Contractor              |
| Address                              |
| Telephone number                     |
| Warranty Contact                     |
| Address                              |
| Telephone number                     |
| Warranty response time priority code |

#### **USE OF ROADWAYS**

For hauling, use only established public roads and roads on Medical Center property and, when authorized by the COR, such temporary roads which are necessary in the performance of contract work. Temporary roads shall be constructed and restoration performed by the Contractor at Contractor's expense. When necessary to cross curbing, sidewalks, or similar construction, they must be protected by well-constructed bridges.



## **TEMPORARY USE OF EXISTING ELEVATORS**

Use of existing elevators for handling building materials and Contractor's personnel will be permitted subject to following provisions:

Contractor makes all arrangements with the COR for use of elevators. The COR will ascertain that elevators are in proper condition. Contractor may use Service Elevators in Building Nos. \_\_740\_\_ for exclusive use for daily use between the hours of \_\_0700-2000\_\_, and for special nonrecurring time intervals when permission is granted. Personnel for operating elevators will not be provided by the Department of Veterans Affairs.

Contractor covers and provides maximum protection of following elevator components:

- a) Entrance jambs, heads soffits and threshold plates.
- b) Entrance columns, canopy, return panels and inside surfaces of car enclosure walls.
- c) Finish flooring.

Government will accept hoisting ropes of elevator and rope of each speed governor if they are worn under normal operation. However, if these ropes are damaged by action of foreign matter such as sand, lime, grit, stones, etc., during temporary use, they shall be removed and replaced by new hoisting ropes at the contractors expense.

If brake lining of elevators are excessively worn or damaged during temporary use, they shall be removed and replaced by new brake lining at the contractors expense.

All parts of main controller, starter, relay panel, selector, etc., worn or damaged during temporary use

shall be removed and replaced with new parts at the contractors expense, if recommended by elevator inspector after elevator is released by Contractor.

Place elevator in condition equal, less normal wear, to that existing at time it was placed in service of Contractor as approved by Contracting Officer.

## **TESTS**

As per specification section 23 05 93 the contractor shall provide a written testing and commissioning plan complete with component level, equipment level, sub-system level and system level breakdowns. The plan will provide a schedule and a written sequence of what will be tested, how and what the expected outcome will be. This document will be submitted for approval prior to commencing work. The contractor shall document the results of the approved plan and submit for approval with the as built documentation.

Pre-test mechanical and electrical equipment and systems and make corrections required for proper operation of such systems before requesting final tests. Final test will not be conducted unless pre-tested.

Conduct final tests required in various sections of specifications in presence of an authorized representative of the Contracting Officer. Contractor shall furnish all labor, materials, equipment, instruments, and forms, to conduct and record such tests.

Mechanical and electrical systems shall be balanced, controlled and coordinated. A system is defined as the entire system which must be coordinated to work together during normal operation to produce results for which the system is designed. For example, air conditioning supply air is only one part of entire system which provides

comfort conditions for a building. Other related components are return air, exhaust air, steam, chilled water, refrigerant, hot water, controls and electricity, etc. Another example of a system which involves several components of different disciplines is a boiler installation. Efficient and acceptable boiler operation depends upon the coordination and proper operation of fuel, combustion air, controls, steam, feedwater, condensate and other related components.

All related components as defined above shall be functioning when any system component is tested. Tests shall be completed within a reasonably period of time during which operating and environmental conditions remain reasonably constant and are typical of the design conditions.

Individual test result of any component, where required, will only be accepted when submitted with the test results of related components and of the entire system. //

## **INSTRUCTIONS**

Contractor shall furnish Maintenance and Operating manuals (hard copies and electronic) and verbal instructions when required by the various sections of the specifications and as hereinafter specified.

Manuals: Maintenance and operating manuals and one compact disc (four hard copies and one electronic copy each) for each separate piece of equipment shall be delivered to the COR coincidental with the delivery of the equipment to the job site. Manuals shall be complete, detailed guides for the maintenance and operation of equipment. They shall include complete information necessary for starting, adjusting, maintaining in continuous operation for long periods of time and dismantling and reassembling of the complete units and sub-assembly components. Manuals shall

include an index covering all component parts clearly cross-referenced to diagrams and illustrations.

Illustrations shall include "exploded" views showing and identifying each separate item. Emphasis shall be placed on the use of special tools and instruments. The function of each piece of equipment, component, accessory and control shall be clearly and thoroughly explained. All necessary precautions for the operation of the equipment and the reason for each precaution shall be clearly set forth. Manuals must reference the exact model, style and size of the piece of equipment and system being furnished. Manuals referencing equipment similar to but of a different model, style, and size than that furnished will not be accepted.

Instructions: Contractor shall provide qualified, factory-trained manufacturers' representatives to give detailed training to assigned Department of Veterans Affairs personnel in the operation and complete maintenance for each piece of equipment. All such training will be at the job site. These requirements are more specifically detailed in the various technical sections. Instructions for different items of equipment that are component parts of a complete system, shall be given in an integrated, progressive manner. All instructors for every piece of component equipment in a system shall be available until instructions for all items included in the system have been completed. This is to assure proper instruction in the operation of inter-related systems. All instruction periods shall be at such times as scheduled by the COR and shall be considered concluded only when the COR is satisfied in regard to complete and thorough coverage. The contractor shall submit a course outline with associated material to the COR for review and approval prior to scheduling

training to ensure the subject matter covers the expectations of the VA and the contractual requirements. The Department of Veterans Affairs reserves the right to request the removal of, and substitution for, any instructor who, in the opinion of the COR, does not demonstrate sufficient qualifications in accordance with requirements for instructors above.

#### **PHOTOGRAPHIC DOCUMENTATION**

During the construction period through completion, provide photographic documentation of construction progress and at selected milestones including electronic indexing, navigation, storage and remote access to the documentation, as per these specifications. The commercial photographer or the subcontractor used for this work shall meet the following qualifications:

Demonstrable minimum experience of three (3) years in operation providing documentation and advanced indexing/navigation systems including a representative portfolio of construction projects of similar type, size, duration and complexity as the Project.

Demonstrable ability to service projects throughout North America, which shall be demonstrated by a representative portfolio of active projects of similar type, size, duration and complexity as the Project.

Photographic documentation elements:

Each digital image shall be taken with a professional grade camera with minimum size of 6 megapixels (MP) capable of producing 200x250mm (8 x 10 inch) prints with a minimum of 2272 x 1704 pixels and 400x500mm (16 x 20 inch) prints with a minimum 2592 x 1944 pixels.

Indexing and navigation system shall utilize actual AUTOCAD construction drawings, making such drawings interactive on an on-line interface. For all documentation referenced herein, indexing and navigation must be organized by both time (date-stamped) and location throughout the project.

Documentation shall combine indexing and navigation system with inspection-grade digital photography designed to capture actual conditions throughout construction and at critical milestones. Documentation shall be accessible on-line through use of an internet connection. Documentation shall allow for secure multiple-user access, simultaneously, on-line.

Before construction, the building pad, adjacent streets, roadways, parkways, driveways, curbs, sidewalks, landscaping, adjacent utilities and adjacent structures surrounding the building pad and site shall be documented. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings. If site work or pad preparation is extensive, this documentation may be required immediately before construction and at several pre-determined intervals before building work commences.

Construction progress for all trades shall be tracked at pre-determined intervals, but not less than once every thirty (30) calendar days ("Progressions"). Progression documentation shall track both the exterior and interior construction of the building. Exterior Progressions shall track 360 degrees around the site and each building. Interior Progressions shall track interior

improvements beginning when stud work commences and continuing until Project completion.

As-built condition of pre-foundation utilities and site utilities shall be documented prior to pouring footers, placing concrete and/or backfilling. This process shall include all underground and in-slab utilities within the building(s) envelope(s) and utility runs in the immediate vicinity of the building(s) envelope(s). This may also include utilities enclosed in slab-on-deck in multi-story buildings. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive site utility plans.

As-built conditions of mechanical, electrical, plumbing and all other systems shall be documented post-inspection and pre-insulation, sheet rock or dry wall installation. This process shall include all finished systems located in the walls and ceilings of all buildings at the Project. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings.

As-built conditions of exterior skin and elevations shall be documented with an increased concentration of digital photographs as directed by the COR in order to capture pre-determined focal points, such as waterproofing, window flashing, radiused steel work, architectural or Exterior Insulation and Finish Systems (EIFS) detailing. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive elevations or elevation details.

As-built finished conditions of the interior of each building including floors, ceilings and walls shall be

documented at certificate of occupancy or equivalent, or just prior to occupancy, or both, as directed by the COR. Overlapping photographic techniques shall be used to ensure maximum coverage. Indexing and navigation accomplished through interactive architectural drawings.

Miscellaneous events that occur during any Contractor site visit, or events captured by the Department of Veterans Affairs independently, shall be dated, labeled and inserted into a Section in the navigation structure entitled "Slideshows," allowing this information to be stored in the same "place" as the formal scope.

Customizable project-specific digital photographic documentation of other details or milestones. Indexing and navigation accomplished through interactive architectural plans.

Monthly (29 max) exterior progressions (360 degrees around the project) and slideshows (all elevations and building envelope). The slideshows allow for the inclusion of Department of Veterans Affairs pictures, aerial photographs, and timely images which do not fit into any regular monthly photo path.

Weekly (21 Max) Site Progressions - Photographic documentation capturing the project at different stages of construction. These progressions shall capture underground utilities, excavation, grading, backfill, landscaping and road construction throughout the duration of the project.

Regular (8 max) interior progressions of all walls of the entire project to begin at time of substantial framed or as directed by the COR through to completion.

Detailed Exact-Built of all Slabs for all project slab pours just prior to placing concrete or as directed by the COR.



Detailed Interior exact built overlapping photos of the entire building to include documentation of all mechanical, electrical and plumbing systems in every wall and ceiling, to be conducted after rough-ins are complete, just prior to insulation and or drywall, or as directed by COR.

Finished detailed Interior exact built overlapping photos of all walls, ceilings, and floors to be scheduled by COR prior to occupancy.

In event a greater or lesser number of images than specified above are required by the COR, adjustment in contract price will be made in accordance with clause entitled "CHANGES" (FAR 52.243-4).

Images shall be taken by a commercial photographer and must show distinctly, at as large a scale as possible, all parts of work embraced in the picture.

Coordination of photo shoots is accomplished through COR. Contractor shall also attend construction team meetings as necessary. Contractor's operations team shall provide regular updates regarding the status of the documentation, including photo shoots concluded, the availability of new Progressions or Exact-Built viewable on-line and anticipated future shoot dates.

Contractor shall provide all on-line domain/web hosting, security measures, and redundant server back-up of the documentation.

Contractor shall provide technical support related to using the system or service.

Upon completion of the project, final copies of the documentation (the "Permanent Record") with the indexing and navigation system embedded (and active) shall be provided in an electronic media format, typically a DVD or external hard-drive. Permanent Record shall have Building Information Modeling (BIM)

interface capabilities. On-line access terminates upon delivery of the Permanent Record.

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# APPENDIX C

**Table 2- Categories and performance qualification levels of water quality for medical device processing.**

| Water Quality Measurement  | Units                   | Utility Water                      | Critical Water                     | Steam*                             |
|----------------------------|-------------------------|------------------------------------|------------------------------------|------------------------------------|
| pH @ 25 °C:                | pH                      | 6.5 - 9.5                          | 5.0 - 7.5                          | 5.0 - 9.2**                        |
| Total Alkalinity           | mg CaCO <sub>3</sub> /L | <400                               | <8                                 | <8                                 |
| Bacteria                   | CFU/mL                  | <500***                            | <10                                | N/A                                |
| Endotoxin                  | EU/mL                   | N/A***                             | <10                                | N/A                                |
| Total Organic Carbon (TOC) | mg/L (ppm)              | N/A                                | <1.0                               | N/A                                |
| Color and Turbidity        | Visual                  | Colorless, clear, without sediment | Colorless, clear, without sediment | Colorless, clear, without sediment |
| <b>Ionic Contaminants</b>  |                         |                                    |                                    |                                    |
| Aluminum                   | mg/L                    | <0.1                               | <0.1                               | <0.1                               |
| Chloride                   | mg/L                    | <250                               | <1                                 | <1                                 |
| Conductivity               | µS/cm                   | <500                               | <10                                | <10                                |
| Copper                     | mg/L                    | <0.1                               | <0.1                               | <0.1                               |
| Iron                       | mg/L                    | <0.1                               | <0.1                               | <0.1                               |
| Manganese                  | mg/L                    | <0.1                               | <0.1                               | <0.1                               |
| Nitrate                    | mg/L                    | <10                                | <1                                 | <1                                 |
| Phosphate                  | mg/L                    | <5                                 | <1                                 | <1                                 |
| Sulfate                    | mg/L                    | <150                               | <1                                 | <1                                 |
| Silicate                   | mg/L                    | <50                                | <1                                 | <1                                 |
| Total Hardness             | mg CaCO <sub>3</sub> /L | <150****                           | <1                                 | <1                                 |
| Zinc                       | mg/L                    | <0.1                               | <0.1                               | <0.1                               |

**Rationale for Table 2 water quality parameters and other considerations:**

a) **pH:** Water having extremes in pH can cause damage of device surfaces. In addition, an extreme pH can interfere with the efficacy of cleaning agents and disinfectants (see E.2.18). Utility water is expected to have a pH in the range of 6.5 to 9.5. The pH of the water will be dependent on the disinfection chemistry (e.g., chlorine, monochloramine) and water treatment strategy of the local municipality. Potable water is typically in the higher portion of this range. For Critical Water the specification is based upon typical values for the water produced. For boiler-treated steam that travels through an extensive distribution network that can be oxidized (e.g., black iron), acidic pH can result in corrosion of the piping and transfer of rust to the sterilizer load, therefore pH should be controlled to maintain a pH of 7.5 to 9.2. For steam produced by a dedicated generator the pH can be in the range of 5.0 to 7.5 (see EN 285).

b) **Alkalinity:** Total alkalinity is the measure of dissolved solids which are alkaline in nature such as bicarbonate, carbonate, and hydroxide. High alkalinity levels increase the tendency of hard water salts to form scale. High alkalinity in the final rinse water increases potential for concentration of alkaline dissolved solids onto the device during drying. This dried alkaline residue can then be reactivated under the high temperature and humidity conditions of a steam sterilizer causing devices to become stained or corroded. The performance of cleaning agents that are not buffered may be impacted by excess alkalinity. Different types of alkalinity may buffer differently and impact scale type or cleaning agents in different ways.

c) **Bacteria:** Critical Water should have very low levels of bacteria. Although water treatment equipment may produce water that is essentially free from microorganisms, the water distribution system may allow for introduction of low levels of bacteria,

requiring routine monitoring. Levels higher than the bacterial levels in Table 2 may increase the bioburden on the load and can increase the risk of patient infections and pyrogenic reaction due to increased endotoxin levels generated by certain types of bacteria.

d) **Endotoxin:** The limit specified by United States Pharmacopeia (USP) for transfusion and infusion assemblies and similar medical devices (USP <161>) is 20 EU remaining on its surface. Critical Water can be derived by a process that would be expected to remove endotoxin. While treatment can reduce endotoxin, some endotoxin might be detected because of passage of the water through tubing that is not endotoxin-free. If the final rinse of a medical device is done with Critical Water containing less than 10 EU/mL, the amount of endotoxin left on the device would be expected to be only a fraction of that amount so the device would have far less than the allowed limit of 20 EU remaining on its surface. Because Utility Water is not produced by a process that removes endotoxin and it is not used for the final rinse of critical devices that contact the patient's bloodstream, endotoxin testing of such water is unnecessary unless it is used as the rinse water after high-level disinfection of medical devices because this rinsing is after the antimicrobial process.

NOTE Adverse effects have been demonstrated when 1 to 4 ng/kg of purified endotoxin (1 ng/kg equals 2 EU/kg) are injected [86]. However, the threshold amount of endotoxin in rinse water that would lead to sufficient residuals on a reprocessed medical device to cause an adverse patient reaction is not clearly defined in the literature. There are no published data or guidelines for endotoxin levels in water used to reprocess medical devices. The 10 EU/mL value described here was adopted by AAMI WG95 committee consensus on the grounds that it was thought unlikely to result in sufficient residual endotoxin on a device to cause an adverse patient reaction.

e) **Total organic carbon (TOC):** Residual TOC should be minimized due to the unknown nature of the TOC and its potential effect on patient tissue and mucosa. The TOC value for Critical Water is based on the expected efficacy of carbon filtration, RO, or distillation in removing organic material and has been defined as less than 1.0 mg/L (ppm). Utility Water would be expected to have total organic carbon levels that are not significantly different than tap or untreated water. Water with high levels of TOC can impact efficacy of UV lamps at controlling microbial growth in water storage systems.

NOTE No studies have been published that establish how much organic material is needed to interfere with disinfection and sterilization processes.

f) **Color and turbidity:** EPA drinking water standards require water to be colorless and clear and contain no sediment. Color and residues are indicative of contaminants and increase the potential to leave contaminants on medical devices during processing. Contaminants left behind on medical devices during certain parts of the processing steps can both stain and/or interfere with subsequent processes. The same criteria for color have been applied to both categories.

### 6.3.1 Ionic characteristics

a) **Conductivity:** this is a measure of the total amount of electrically charged impurities (ions) that are present in the water sample. It is measured in terms of electrical current over a distance with units of  $\mu\text{S}/\text{cm}$ . It is a broad measurement of the impurities in the sample and is used as a proxy for Total Dissolved Solids. It gives an indication of the general purity of the sample. High conductivity indicates the presence of large amounts of charged ions, some of which can result in corrosion or staining of instruments.

b) **Chloride:** When chloride ions are present in sufficient concentrations they can result in corrosion of stainless steel or soft metals.

c) **Nitrate:** residual nitrates can be very reactive with other compounds. Residual amounts of nitrates on medical device surfaces may affect subsequent processes.

d) **Phosphate:** phosphates can enter water as run off from agricultural uses and can contribute to scale formation, leave residues on instrumentation, and interact with cleaning chemicals

**Sulfate:** sulfates can form scale with positively charged ions. Routinely monitored water treatment systems with increasing levels of sulfate can be indicative of resin bed issues.

f) **Silicate:** silicates can form hard to remove residues, especially if allowed to remain on instruments during subsequent steps of processing involving heat.

g) **Iron:** iron ions can deposit on stainless steel surfaces and cause corrosion. It can also result in rust/corrosion that can manifest in a variety of colors (e.g., brown, black, etc.).

h) **Copper:** copper ions can deposit on surfaces of items being processed and cause green staining and/or corrosion.

i) **Manganese:** manganese ions can deposit on surfaces of items being processed and cause black staining and/or corrosion.

j) **Aluminum:** aluminum ions can deposit on surfaces of items being processed and cause staining and/or corrosion.

k) **Zinc:** zinc ions can deposit on surfaces of items being processed and cause staining and/or corrosion.

NOTE Metal ions (e.g., iron, copper, zinc, etc.) can also contribute to hardness values and deposit on metal surfaces resulting in galvanic corrosion.

l) **Water hardness:** Water hardness is a critical parameter to control as it affects residues on devices. Tap water is generally considered "hard" if it has calcium and magnesium levels higher than 150 ppm; therefore, this level is given as the upper limit for Utility Water. Avoiding excess hardness prolongs the life of washer-disinfectors; reduces the risk of hard-water deposits on medical devices during processing and helps ensure the efficacy of cleaning agents. If hardness is greater than 150 mg/L(ppm), water softening is recommended unless used for washing where the cleaning chemistry is capable of handling higher levels of hardness or metal ions in solution. For Critical Water, the treatment process would be expected to remove contaminants and produce water that has calcium and magnesium levels of less than 1 ppm. Using Critical Water for the final rinse of critical medical devices reduces the likelihood of hardwater deposits.

There are performance qualification specifications for water quality defined in Table 2. For ongoing monitoring of the quality, a reduced set of specifications with greater frequency (Table 4), allows for setting of a baseline of expected quality parameters. The values expressed in Table 4 (i.e., action levels) are not to be exceeded, however trending baseline data and setting an alert level can give early indication of the water quality trend. The baseline data can be used for indicating variations that may impact the water quality and setting of appropriate alert levels.

The alert levels can be identified using a method determined by the facility's multidisciplinary team or can be calculated using the example provided in Clause G.1.2.

An alert should be triggered if a sample results in data outside of the range defined by the upper control limit (UCL) and the lower control limit (LCL).

When data is outside the upper or lower alert levels, it is an excursion, a deviation from a regular pattern, path, or level of operation. Any excursions outside of those typically seen values (within action levels) can be identified as warning of potential issues. The monitoring program is an opportunity to address potential problematic issues prior to failing of specifications and can help anticipate repair or replacement of water treatment equipment or processes.

An excursion (e.g., data point outside alert level) should prompt investigation in order to find the assignable causes and associated risks and then mitigate, where possible. These causes may indicate a change in conditions (e.g., seasonal, construction, repair).

## **11 Continuous quality improvement**

### **11.1 Introduction**

This section identifies performance measures and process monitors that can be used for continuous quality improvement (CQI) programs. CQI programs are recognized as an effective means of improving the performance of any process. For water quality, a CQI program encompasses the entire process of decontamination through sterilization. The quality of the water is important in each stage and needs to be monitored at various times in the process.

## **11.2 Quality process**

Procedures for determining water quality should be based on a documented quality process that measures objective performance criteria (e.g., pH, hardness, purity, temperature) that have a direct effect on the outcome. This quality process should be developed in conjunction with personnel from other areas (e.g., facility engineering, the sterile processing department, infection prevention and control) and integrated into the overall quality process in the health care facility. Variables in the system may be controlled to achieve assurance of product quality and process efficacy. Monitoring frequency will vary, depending on the quality improvement goals, the health care facility policies and procedures for the handling of unfavorable or unplanned events, and the type of process variable.

An investigation should be conducted for any problem relating to any aspect of water quality. Problems and updates should be reported to those involved with medical device processing.

There should be a planned, systematic, and ongoing process for verifying compliance with procedures. Quality processes can be enhanced by audits that are conducted on a regular basis. The information from these activities should be summarized and made available to individuals, groups, or teams as needed.

Table 5 and Table 6 describe the water quality monitoring that shall be performed as part of a quality improvement process within their area. Continuous quality improvement involves conducting risk analyses.

## **Water treatment systems maintenance**

### **12.1 General considerations**

Proper maintenance of a well designed water treatment system should follow the original equipment manufacturers (OEM) recommendations. Improper maintenance can lead to equipment performance issues, water system component failure, and/or improper water quality levels that do not meet the guidelines of this standard. Bacterial growth, increased endotoxin levels, and poor final water quality are indicative of a poorly maintained water treatment and delivery system.

Loop systems and tanks (see Annex G.4) should be disinfected monthly as a best practice.

A regular system maintenance schedule should be followed to meet OEM recommendations including expendable components based on site specific water conditions, at a minimum, and indicators should be defined for each piece of equipment in the water treatment system.

It is recommended that a daily checklist be used to monitor basic system parameters, such as loop final water quality and critical OEM directed readings, which may include levels of expendable items that need to be monitored.

Testing of at least the loop out and the loop return should be done no less frequently than monthly to verify bacterial levels and endotoxin levels.

### **12.2 Serviceability of system**

Systems shall be designed with clear guidance on system controls that are easy to follow and reduce errors. The system should be allotted adequate room for installation that will allow access to all components for maintenance and allow adequate access around the equipment. Space should be allowed in between components and the ends of the system to allow for replacement of expendable items and repairs. Systems should meet all local building codes.

Reverse osmosis systems should be set up to permit disinfection and allow for low and high pH chemical cleanings for ongoing maintenance.

Systems should be on a fully recirculating loop to allow for full and effective disinfection of the system piping and have a fully drainable tank for complete and faster disinfection. Existing direct feed systems should be updated to recirculating systems, when possible, to allow for effective disinfection. Until such time that the loop is installed, the direct feed may be required to be monitored and disinfected more frequently to reduce the risk of bacterial growth and/or endotoxin accumulation. All newly designed and installed systems shall be on a fully recirculating loop. Sample ports shall be placed at the loop out after final filtration as well as right before re-entry into the storage tank. Where feasible, an accessible sample port should be placed before individual points of use (e.g., immediately before the inlet to the washer-disinfector).

## **13 Special considerations**

### **13.1 Post construction or extended shutdown**

Typically performed by facilities engineering personnel:

- ☐ Reverse osmosis system/ultra filtration should be flushed. Flushing should be run every 24-hours for a minimum of 30-minutes or follow OEM recommendations.
- ☐ Pretreatment apparatus or filters should be scheduled to backwash or regenerate every 48-hours maximum.
- ☐ Pretreatment filter(s) should be changed out prior to putting into service.
- ☐ Water treatment equipment should be disinfected prior to putting into service.
- ☐ Flush Critical Water systems, by running water for two system volumes post disinfection. Contact the processing equipment manufacturer for guidance regarding how to flush stagnant water from equipment.
- ☐ Remove aerators and flush hot and cold sink water for a minimum of 5-minutes or until visibly clear and free of particles/debris.
- ☐ Total bacteria and endotoxin tests should be performed prior to putting into service.
- ☐ After construction, or an extended period of disuse, Critical Water systems should be flushed to remove stagnant water prior to sampling. Bacterial testing and endotoxin testing should be performed on the loop.
- ☐ Upon receipt of total bacteria and endotoxin tests with acceptable results, the unit may be placed into service.

### **13.2 Extended boil water alerts and steps to take after alerts are lifted**

- ☐ Once the boil water alert has been lifted, flush all equipment (e.g., water treatment equipment/loop) with a disinfectant that is compatible with piping materials and equipment and restart. Also, flush feedwater source (prior to treatment equipment).
  - o After disinfectant application is complete, flush for at least five minutes and until residual disinfectant is detected at feedwater source prior to treatment.
  - o If chlorine cannot be detected at point-of-use after flushing, supplemental disinfection may need to be considered (drinking water regulations will apply, by performing supplement disinfection).
- ☐ Drain, disinfect, flush, and refill water storage tank.
- ☐ Change pretreatment filters, backwash carbon tanks, regenerate softener, clean and disinfect RO membranes, disinfect distribution loop.
- ☐ If steam sterilizers are used in processing, check the steam system.
  - o Check water quality used for steam generation.
  - o Check with manufacturer for written instructions for bringing the system back online following a contamination event.
- ☐ If using deionization tanks these should be replaced with new deionization tanks prior to going back into service.
- ☐ The loop final filter should be replaced with a new absolute rated final filter.

### **13.3 Interruptions in service**

If an interruption of service is from the utility feedwater source or a building system, shut down and restart.

- ☐ Apply disinfectant and flush for at least five minutes and until residual disinfectant is detected at feedwater source prior to treatment. Disconnect the feed to the unit and flush the feed source directly to the drain.
- ☐ If disinfectant cannot be detected at point-of-use after flushing one may need to consider supplemental disinfection (drinking water regulations will apply, by performing supplemental disinfection the health care facility would now be considered a small public water system).

If the system cannot be disconnected from the feed source and directed to drain:

- Change pretreatment filters, backwash carbon tanks, regenerate softener, clean and disinfect RO membranes, and disinfect distribution loop.

- If steam sterilizers are used in processing, check the steam system. o Check water quality used for steam generation.

- o Check with manufacturer for instructions for bringing system back online following a contamination event.

- If using deionization tanks these should be replaced with new deionization tanks prior to going back into service.

- The loop final filter should be replaced with a new Absolute Rated Final Filter.

### **13.4 System repair, modification and/or routine maintenance on a Critical Water production, storage, and distribution system**

A Critical Water system is or should be designed to deliver water meeting all specified parameters to the points of use. See Clause 8.3 for details of Critical Water system design and Table 2 for the specified parameters.

Routine maintenance, repair, or system expansion can require that the Critical Water system be opened to the atmosphere, which exposes the piping and water in the system to ambient airborne microbiological flora. In the interests of patient safety, these should be assumed to be pathogenic. Thus, any time piping is replaced, or system modification is completed on a Critical Water system, the system piping and storage should be disinfected.

Disinfection should be performed after installation, expansion, major repairs, non-compliant bacterial levels, or other changes to the system (e.g., nearby building construction or water supply interruptions). In addition, disinfection should ideally be completed for prophylactic purposes, even if no non-compliant bacterial or endotoxin results have been observed (See Annex G).

Specific instances in normal maintenance in which system disinfection is required include:

#### **13.4.1 Pretreatment equipment**

Examples: cartridge particulate filtration, multimedia filtration, softener, carbon filter, sodium bisulfite injection system, and anti-scalant injection system

This equipment typically does not require regular disinfection. However, if testing indicates non-compliant bacterial levels at the sample port immediately downstream of the pretreatment equipment, the equipment should be disinfected according to material compatibility. If testing still indicates non-compliant bacterial levels, the filtration media should be replaced. Non-compliant levels are defined here as a bioburden level of too numerous to count.

#### **13.4.2 Primary treatment equipment**

EXAMPLE: reverse osmosis (RO), deionization (DI), and electrodeionization (EDI)

##### **13.4.2.1 Reverse Osmosis (RO) unit**

RO units should be cleaned, using high and low pH cleaning chemicals, when permeate (product water) indicates membrane fouling and degradation of performance. This will be dependent upon the quality of the feedwater to the RO unit, as well as the volume of water filtered by the RO. Therefore, frequency should be determined by the specific use profile of the sterile processing department and when data indicates cleaning is required. At a minimum, this should be performed at least annually.

RO units should also be disinfected using a compatible disinfection method at least annually or if testing indicates non-compliant bacterial levels at the sample port immediately downstream of the RO unit.

##### **13.4.2.2 Deionization (DI) exchange tanks**

DI exchange tanks are typically not compatible with the disinfection methods.

Therefore, the DI exchange tanks should be exchanged with replacement tanks when testing indicates non-compliant bacterial levels at the sample port immediately downstream of the DI tanks and there is not a bacteriostatic filter downstream of the DI tanks.

##### **13.4.2.3 Electrode ionization (EDI) unit**

EDI units should be disinfected when testing indicates non-compliant bacterial levels at the sample port immediately downstream of the EDI unit.

##### **13.4.3 Storage tanks and distribution equipment and piping**



Examples: storage tanks, distribution pump(s), UV disinfectant, bacteriostatic filtration, and distribution loop piping

High purity water storage, distribution equipment and piping should be disinfected using a compatible disinfection method if testing indicates non-compliant bacterial levels at the sample port immediately downstream of the final treatment technology, typically the bacteriostatic filters, at the inlet of the distribution loop, typically the bacteriostatic filters and on the sample port returning. If bacteriostatic filters are not compatible with disinfection methods (see Annex G), the filters must be replaced with new filters. Bacteriostatic filters should be replaced if the pressure difference (delta pressure, or  $\Delta P$ ) across the filter is 10 PSI or greater, or downstream testing indicates biological breakthrough and/or loop biofouling. Localized disinfection is required for areas that can be bypassed for normal maintenance. For example, the bacteriostatic filter if there is a bypass permitting its replacement without exposing the entire water system to ambient atmospheric conditions. Bypasses should be designed to ensure that stagnant water within them does not allow for bacterial growth or biofilm formation. If the bacteriostatic filter at the inlet of the distribution loop cannot be isolated during replacement, the distribution loop should be disinfected after the filters are replaced.

Storage tanks, distribution equipment, and distribution loop piping should be disinfected after the replacement of in-line probes and sensors. Storage tanks, distribution equipment, and distribution loop piping should be disinfected after service and/or replacement of recirculation pumps and associated piping. Storage tanks, distribution equipment, and distribution loop piping should be disinfected after a shutdown of a maximum duration determined by the multidisciplinary water management team, or when post-shutdown testing, if performed, indicates non-compliant bacterial levels.

## **Microbiological quality – bacterial endotoxin test**

### **H.4.1 *Limulus* amoebocyte lysate (LAL)**

Testing for bacterial endotoxin is generally performed by means of the *Limulus* amoebocyte lysate (LAL) assay. Two basic types of LAL assay can be performed: the kinetic assay, which is available in a colorimetric or turbidimetric format, and the gel-clot assay. Alternatively, validated test methods to detect endotoxin or pyrogens may be used. The LAL assay is described in ANSI/AAMI ST72 [7]. A positive control and a negative control should be run with each assay. The limits for endotoxin levels are given in Table 2.

## **How to perform water sampling**

### **H.5.1 Sample collection**

Critical Water samples, or other types of water when indicated, should be collected from several locations within the distribution system to provide a representative microbial water quality of the system. In general, samples should be collected at the beginning and end of the water distribution loop. A point of use in the distribution loop is a location where water enters equipment utilized to process medical devices. Where any storage tank is used, water should be sampled at the end of the purification process and the exit of the storage tank.

Samples should always be collected before disinfection of the water treatment system. If repeat cultures are performed after the system has been chemically disinfected (e.g., with hydrogen peroxide, chlorine, or peracetic acid), the system should be flushed completely before samples are collected. Before samples are collected, the storage tanks and distribution system should be drained and flushed until residual disinfectant is no longer detected. For new systems, the water should be cultured weekly until an established pattern has been determined. For established systems, the water can be cultured monthly unless a greater frequency is dictated by historical data at a given institution.

Water samples should be collected directly from outlet taps situated in different parts of the water distribution system. In general, the sample taps should be opened, and the water allowed to run for at least 60 seconds before a sample is aseptically collected in a sterile, endotoxin-free container. The collection container should not be made of polypropylene, because polypropylene has been shown to inhibit endotoxin detection [78]. At least 100 to 300 mL of water should be collected. Sample taps should be treated with a general disinfectant (e.g., alcohol) prior to sampling and be completely dried to prevent contamination of the collected sample disinfectant

residual from the sample tap. Bleach or other disinfectant solutions should never be used. Bacterial replication is prevented by holding the sample on ice, or at 4 °C, until it is delivered to the laboratory for testing.

#### **H.5.2 Water sampling technique**

**Sampling Consistency:** Sampling should be consistent in procedure and sampling locations to ensure the accurate monitoring of water quality.

**Flushing:** Sample points should be flushed commensurate to the size of the water system and distribution loop. Biofilms can develop throughout the water system, including infrequently used valves or downstream connectors between the distribution loop and equipment. Flushing these areas with a high flow velocity across these surfaces can shear off fragile tops of biofilms possibility growing in these areas. A standardized flush before sampling can limit the microbial contribution from the sampling points.

**Sampling containers:** The appropriate sampling container should be used to collect the water sample. The container should be of the required quality not to introduce contamination into the test sample. For example, samples used in the sampling for bacterial endotoxin should be of the appropriate quality (e.g., pyrogen-free).

**Sampling technique:** Valves/taps/hoses used for sampling should be of the appropriate design to not contaminate the sample. For example, sampling locations do not provide a stagnant water environment when not in use and is accessible to the distribution loop disinfection process. Prior to sampling the sampling surfaces (e.g., valves, hoses) should be disinfected both externally and internally to reduce microbial contamination. Aseptic technique should be used for water collection.

#### **Results exceeding specification**

Microbial quality levels detected, and exceeding specification should lead to a repetition of the disinfection and sampling at the location to confirm the results. Upon confirmation of the exceedance, immediate action is required to reduce the microbial contamination. The distribution loop should be disinfected using methods described in Annex E to ensure the microbial contamination is not being introduced through the distribution system. Automated instrument reprocessors (e.g., washer-disinfectors, washer-decontaminators, AERs) should be flushed with 1,000 ppm of a chlorine-releasing agent after the first instance of persistent unacceptable microbial levels is detected [88]. Other agents that could be considered for flushing include chlorine dioxide, peracetic acid, hydrogen peroxide, and super oxidized water. Before the unit is flushed with any of these agents, it is important to confirm with the reprocessor manufacturer that the components of the machine are compatible with the disinfectant to be used in its intended concentration. The levels should be monitored weekly by repeat testing until two successive counts show that baseline levels have been achieved.

If the microbial levels continue to exceed acceptable limits despite disinfection activities, a second system disinfection, combined with replacement of as much of the pipework as is practical, should be undertaken to remove biofilm. For the second round of disinfection, a higher concentration of chlorine-releasing agent (e.g., 10,000 ppm) or an alternative disinfectant should be used [88].

If contamination problems persist despite all efforts, the main water source should be investigated, and local authorities asked for input. This approach is based on the action plan outlined by the Working Party of the Hospital Infection Society and the Public Health Laboratory Service [88].

## APPENIX D

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